Passive Fault Tolerance through Tension-to-Thrust Feed-Forward: Hybrid Input-to-State Stability for Decentralized Multi-UAV Slung-Load Transport under Abrupt Cable Severance

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***Abstract*—Cooperative aerial manipulation under unannounced cable severance demands a fault response on the payload-pendulum timescale ($\approx 2\,\mathrm{s}$), well below the latency of any detection–classification–reconfiguration pipeline. For this class of self-announcing structural transitions, we show that a single decentralized feed-forward identity, $T_i^{\mathrm{ff}} = T_i$, equating each drone's altitude thrust feed-forward to its own measured rope tension, supplies the dominant passive fault-response mechanism. The identity supplies the bounded-jump mechanism required by the hybrid practical input-to-state stability (ISS) certificate, enabling recovery across unannounced severance events without any estimator, detector, or peer communication.**

We establish three results. First, a taut-cable reduction theorem with explicit, numerically verified constants certifies an $O(\delta + \eta_{\max})$ admissibility bound on a slack-excursion-bounded domain, replacing the distributed Kelvin–Voigt rope dynamics with a lumped reduced-order plant. Second, a conditional hybrid practical-ISS theorem yields an explicit recovery envelope with closed-form Lyapunov decay rate λ and per-fault-cycle contraction $\rho \in (0, 1)$; the fault-jump bound $\chi(\Delta_f)$ is explicit in structure and bounded in terms of κ_V, whose numerical calibration is a stated follow-on. Third, a scalar admissible-window rule for an optional L_1 adaptive altitude augmentation derives two necessary conditions — discretization stability and bandwidth coverage — that together place the canonical gain inside a certifiable design window, reducing reliance on empirical tuning. Because each drone's closed loop is identical pre- and post-fault and contains no peer-coupled term, the architecture admits a sequence of $F \leq N-2$ unannounced severances by induction on the fault counter, subject only to the actuator-margin envelope $m_L g/(N-F) \leq \kappa_{\mathrm{act}} f_{\max}$.

In Drake multibody simulation across a pre-declared six-variant capability sequence and four ablation campaigns at $N = 5$, $m_L = 10\,\mathrm{kg}$, and $4\,\mathrm{m\,s^{-1}}$ Dryden wind, the controller absorbs single and dual unannounced severances within one payload-pendulum period. The feed-forward ablation isolates the identity as the causal mechanism: disabling it inflates cruise RMSE by 34–$39\,\%$ and post-fault altitude sag by 3.6–$4.0\times$. Two null findings — that the model-predictive-control tension-ceiling layer and the post-fault reshape supervisor remain structurally inactive on the demonstrated trajectory family — support the interpretation that the baseline passive mechanism, rather than conventional active fault-tolerance machinery, explains the observed recovery.

***Index Terms*—Cooperative aerial manipulation, cable-suspended payload transport, decentralized fault-tolerant control, passive fault tolerance, hybrid practical input-to-state stability, Lyapunov contraction analysis, L_1 adaptive augmentation, receding-horizon model predictive control, anti-swing control, structural-transition systems.**

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I. Introduction

We are interested in cooperative aerial transport, where N multirotor uncrewed aerial vehicles (UAVs) lift a common payload through flexible cables [1]–[4]. Cable severance—from abrasion, hardware fatigue, or obstacle contact—is a structural fault: it discretely reduces the constraint set, redistributes load across surviving cables within one payload-pendulum period ($\tau_{\mathrm{pend}} = 2.24\,\mathrm{s}$ at $L = 1.25\,\mathrm{m}$), and requires a compensatory response within that window.

Unilateral coupling, hybrid constraint-cardinality transitions at severance events [5], [6], and strict information locality together rule out classical cooperative and slung-load architectures: any admissible controller must deliver post-fault response without detection, annunciation, or reconfiguration and must remain analytically valid across structural transitions.

This paper shows that, for this plant class, a single feed-forward identity in the per-drone controller can provide the dominant passive fault-response mechanism, rather than relying on detection and reconfiguration: drone i's altitude thrust feed-forward is set equal to its own measured rope tension,

$$T_i^{\mathrm{ff}}(t) = T_i(t),$$

formally given in (36) in Section III. While the use of measured interaction forces in control is not new, we show that, in this setting, this identity has a non-obvious consequence: it induces a certifiable stabilizing response across structural fault transitions on the plant's native timescale, without detection or reconfiguration.

When a peer's cable severs, the load redistributes within one payload-pendulum period (≈ 2 s) and the surviving drone's tension rises. Identity (36) converts that rise at the next control tick into a proportional thrust increment, absorbing the new load at the plant's native rate without detection or communication. The mechanism sits inside a conventional cascade—an outer-loop PD controller with anti-swing slot shift, a single-step convex QP on the tilt-and-thrust envelope, and a geometric attitude loop—with (36) as the sole nonstandard component.

We develop the consequences of (36) along three axes. The primary contribution (C2, Section IV) is a hybrid practical-ISS theorem: under a single-active-regime QP doctrine and a spaced-fault dwell hypothesis $\tau_d \geq \tau_{\mathrm{pend}}$, the closed-loop tracking error admits a closed-form recovery envelope with Lyapunov decay rate λ and per-fault-cycle contraction $\rho \in (0, 1)$, conditional on bounded slack excursions, spaced fault events, and a single-active-regime QP—all audited in the simulation campaign. The strict $\rho < 1$ is achieved by

(36): the dwell hypothesis is the minimum time over which $T_i^{\mathrm{ff}} = T_i$ restores the Lyapunov function below its pre-fault level (Lemma 5). The first secondary contribution (C1, Section II-C) is a taut-cable reduction theorem: on a slack-excursion-bounded domain parameterized by the timescale ratio δ and duty-cycle bound η, an explicit first-order error budget justifies replacing the distributed rope dynamics with a lumped effective stiffness; all admissibility gates are met with margin by all six missions (Table VI). The third contribution (C3, Section V) is a scalar admissible-window design rule for the L_1-adaptive altitude augmentation: two necessary conditions — discretization stability and bandwidth coverage — derived from the same scalar discrete pole together place the canonical gain Γ inside a certifiable interval $[\Gamma_{\min}, \Gamma^\star]$, reducing reliance on empirical tuning. This is an auxiliary engineering rule, not a full discrete-time L_1 stability certificate for the complete interconnection.

The proposed controller is validated in high-fidelity Drake multibody simulation through a pre-declared six-variant capability demonstration and four targeted stress campaigns on a five-drone, $10\,\mathrm{kg}$-payload, $4\,\mathrm{m\,s^{-1}}$ Dryden-wind scenario under single and compound cable-severance faults. The feed-forward ablation campaign isolates identity (36) as the causal mechanism: disabling it materially inflates cruise tracking error and post-fault altitude sag (Table IX). Two null findings are reported alongside: in these regimes, a composable extension layer is structurally inactive because the baseline already satisfies the relevant constraint. The architecture admits a sequence of $F \leq N-2$ unannounced severances under the actuator-margin envelope (Section II-B); the demonstrated $N = 5$, $m_L = 10\,\mathrm{kg}$ admits $F \leq 2$.

The contribution identifies a stability mechanism for structural transitions that self-announce through actuator-side measurements, not a feature combination on a five-drone case study. Four architectural consequences follow from (36): *locality* ($\mathcal{I}_i$ (13) contains no peer state, fault flag, or shared scheduler); *passivity* (no detector, classifier, or reassignment rule); *robustness to $F \leq N-2$ severances* by induction under the actuator-margin envelope (26); and *team-size independence* (the per-drone law carries no N structurally; dependence concentrates in the actuator-margin envelope). These properties are audited in Section VII-B; the two null findings (Remarks 5 and 6) confirm the baseline mechanism explains the observed behavior.

The remainder of the paper is organized as follows. Section II formalizes the plant, fault model, and taut-cable reduction (C1); Section III presents the three-layer decentralized cascade; Section IV establishes the hybrid practical-ISS theorem (C2) through six supporting lemmas; and Section V develops the L_1 admissible-window bound (C3), MPC tension ceiling, and reshape supervisor, each preserving the recovery envelope on its admissibility domain.

A. Related Work

Related work falls into four categories: centralized and distributed cooperative transport, geometric tracking and slung-load control, L_1 adaptive control, and fault-tolerant control.

a) Centralized and distributed cooperative transport.: Force-allocation architectures [1], [2], [4] and distributed predictive variants [7]–[10] share a synchronous global-state requirement: the per-tick communication dependency is precisely what a cable-severance transition breaks at the actuator-tick rate, and none of these architectures accounts for cable severance in its stability argument.

b) Geometric tracking and slung-load control.: Geometric tracking [11], flatness-based slung-load [12], and $\mathrm{SE}(3)$ payload control [13] assume a fixed constraint set; multi-vehicle extensions [14], [15] reintroduce global-state exchange, collapsing the strict locality the present architecture requires.

c) L_1 adaptive control.: L_1 adaptive control [16], [17] provides matched-disturbance rejection with transient bounds; its extension to cooperative cable-fault settings under strict locality remains open [18].

The fault-tolerant control literature divides into active architectures that reconfigure after diagnosis [19], [20] and passive architectures that absorb faults through robustness [19], [21]. Within cooperative transport, observer-based disturbance estimation with dynamic load reassignment [22] addresses cable failures through detection and reconfiguration, while passive force control [23] and leader–follower force-estimation approaches [24] avoid inter-UAV exchange but presuppose fixed cable cardinality. The present contribution is best read as a different commitment about *how the structural-transition announcement is processed*: the actuator-side tension measurement *is* the announcement, eliminating the detection-classification-reconfiguration timeline as a system-level construct. The novelty claimed here is the stability certification of a decentralized tension-to-thrust mechanism for a structural fault whose effect is directly measured through local cable tension.

II. Problem Statement

We study cooperative aerial transport under unannounced cable-severance faults and formalize the plant, fault model, admissible controller class, reference trajectory, taut-cable reduction, and formal problem.

A. Plant Description and Admissible Disturbance Set

a) Vehicles, payload, and state.: A team of N quadrotor UAVs, indexed $i \in \mathcal{N} \triangleq \{1, \ldots, N\}$, operates in an inertial world frame $\mathcal{W}$. Drone i has mass m_i, diagonal inertia $\boldsymbol{J}_i = \mathrm{diag}(J_x^{(i)}, J_y^{(i)}, J_z^{(i)})$, position $\boldsymbol{p}_i \in \mathbb{R}^3$, velocity $\boldsymbol{v}_i \in \mathbb{R}^3$, attitude $\boldsymbol{R}_i \in \mathrm{SO}(3)$, and angular velocity $\boldsymbol{\omega}_i \in \mathbb{R}^3$. The per-drone state is $x_i \triangleq (\boldsymbol{p}_i, \boldsymbol{v}_i, \boldsymbol{R}_i, \boldsymbol{\omega}_i)$ and the collective drone state is $x_{\mathcal{N}} \triangleq (x_i)_{i=1}^N$. A point-mass payload of mass m_L has position $\boldsymbol{p}_L \in \mathbb{R}^3$, velocity $\boldsymbol{v}_L \in \mathbb{R}^3$, and state $x_L \triangleq (\boldsymbol{p}_L, \boldsymbol{v}_L)$; payload rotation is neglected.

b) Cables and complementarity condition.: Each drone i is connected to the payload by a flexible cable. Cable i exerts a world-frame force $\boldsymbol{T}_i^{\mathcal{W}} \in \mathbb{R}^3$ on the payload-side attachment point and an equal and opposite reaction on the drone-side

attachment point. The cable transmits tensile force only when taut; its scalar tension is

$$T_i \triangleq \left\|\boldsymbol{T}_i^{\mathcal{W}}\right\| \geq 0, \tag{1}$$

with the unilateral constraint $T_i = 0$ whenever the cable is slack. The complementarity condition

$$T_i \geq 0, \quad \ell_i - L \leq 0, \quad T_i\left(\ell_i - L\right) = 0, \tag{2}$$

where $\ell_i = \|\boldsymbol{p}_{a,i} - \boldsymbol{p}_{b,i}\|$ is the cable's instantaneous length, $\boldsymbol{p}_{a,i}$ and $\boldsymbol{p}_{b,i}$ are the drone- and payload-side attachment points, and L is the rest length, captures the unilateral tensile constraint for an idealized inextensible cable. In the simulator each cable is a Kelvin–Voigt spring ($L = 1.25\,\mathrm{m}$, $k_s = 25\,000\,\mathrm{N\,m^{-1}}$); the elastic correction to the inextensible model is $O(\delta)$ and is absorbed in Theorem 1.

c) Equations of motion.: The translational dynamics of drone i are governed by

$$m_i\,\dot{\boldsymbol{v}}_i = f_i\,\boldsymbol{R}_i\,\hat{\boldsymbol{e}}_3 - m_i g\,\hat{\boldsymbol{e}}_3 - \boldsymbol{T}_i^{\mathcal{W}} + \boldsymbol{F}_{\mathrm{w},i}(t), \tag{3}$$

where $f_i \in [f_{\min}, f_{\max}]$ is the scalar thrust, $\hat{\boldsymbol{e}}_3$ is the world-frame vertical unit vector, and $\boldsymbol{F}_{\mathrm{w},i}$ represents an aerodynamic wind disturbance. The rotational dynamics are

$$\boldsymbol{J}_i\,\dot{\boldsymbol{\omega}}_i + \boldsymbol{\omega}_i \times \boldsymbol{J}_i\,\boldsymbol{\omega}_i = \boldsymbol{\tau}_i, \qquad \dot{\boldsymbol{R}}_i = \boldsymbol{R}_i\,[\boldsymbol{\omega}_i]_\times, \tag{4}$$

where $\boldsymbol{\tau}_i$ is the body-frame control torque and $[\cdot]_\times$ is the skew-symmetric operator. The payload translational dynamics are

$$m_L\,\dot{\boldsymbol{v}}_L = -m_L g\,\hat{\boldsymbol{e}}_3 + \sum_{i\in\mathcal{S}(t)} \boldsymbol{T}_i^{\mathcal{W}} + \boldsymbol{F}_{\mathrm{w},L}(t), \tag{5}$$

where $\mathcal{S}(t) \subseteq \mathcal{N}$ is the set of surviving (intact) cables at time t and $\boldsymbol{F}_{\mathrm{w},L}$ is the wind disturbance on the payload.

d) Inputs, state space, and admissible disturbance set.: The per-drone control input is

$$\begin{aligned} u_i(t) &\triangleq \left(f_i(t),\ \boldsymbol{\tau}_i(t)\right) \in \mathcal{U}_i \\ \mathcal{U}_i &\triangleq [f_{\min}, f_{\max}] \times [-\tau_{\max}, \tau_{\max}]^3, \end{aligned} \tag{6}$$

and the collective input is $u(t) \triangleq (u_i(t))_{i=1}^N$. The complete plant state is

$$\begin{aligned} x(t) &\triangleq \left(x_{\mathcal{N}}(t),\ x_L(t)\right) \in \mathcal{X} \\ \mathcal{X} &\triangleq \left(\mathbb{R}^3 \times \mathbb{R}^3 \times \mathrm{SO}(3) \times \mathbb{R}^3\right)^N \times \mathbb{R}^3 \times \mathbb{R}^3. \end{aligned} \tag{7}$$

Collecting the wind disturbances into $d(t) \triangleq (\boldsymbol{F}_{\mathrm{w},1}(t), \ldots, \boldsymbol{F}_{\mathrm{w},N}(t), \boldsymbol{F}_{\mathrm{w},L}(t))$, the admissible disturbance set is

$$\begin{aligned} \mathcal{D} \triangleq \Big\{ d(\cdot) : \|\boldsymbol{F}_{\mathrm{w},i}(t)\| \leq W_{\max},\ \|\boldsymbol{F}_{\mathrm{w},L}(t)\| \leq W_{\max}, \\ \forall t \geq 0,\ \forall i \in \mathcal{N} \Big\}, \end{aligned} \tag{8}$$

where $W_{\max} = 1\,\mathrm{N}$. The Dryden realization used in V2–V6 remains inside this bound (drag and frontal-area parameters in Table IV).

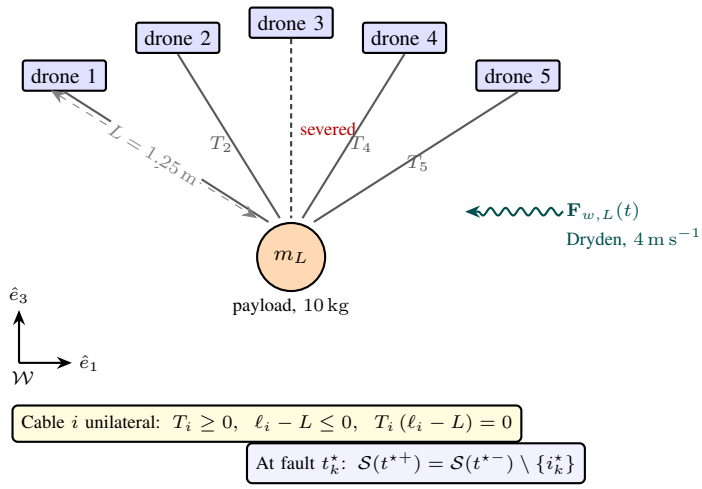


Fig. 1. Cooperative transport configuration. Five quadrotors carry a 10 kg point-mass payload through unilateral cables of rest length $L = 1.25\,\mathrm{m}$. Each drone i measures only its own state, its own scalar tension T_i, and the locally observed payload velocity $\boldsymbol{v}_L$; no peer drone or peer-cable signal is exchanged. The plant is driven by a bounded Dryden wind realisation ($4\,\mathrm{m\,s^{-1}}$ mean) and an unannounced cable-severance fault schedule $\sigma \in \mathcal{F}$ that decrements the survivor set $\mathcal{S}(t)$ at each event $t_k^\star$ (here drone $i^\star$, shown dashed).

B. Fault Model and Admissible Controller Class

a) Structural fault event.: At a time $t_k^\star$, cable $i_k^\star \in \mathcal{N}$ severs instantaneously. Severance permanently removes cable $i_k^\star$ from the constraint set:

$$T_{i_k^\star}(t) = 0 \qquad \forall\, t \geq t_k^\star. \tag{9}$$

The active cable set transitions discretely:

$$\mathcal{S}(t_k^{\star,+}) = \mathcal{S}(t_k^{\star,-}) \setminus \{i_k^\star\}, \tag{10}$$

reducing the cardinality from $|\mathcal{S}|$ to $|\mathcal{S}| - 1$. For an elastic cable, the kinematic state $(\boldsymbol{p}_i, \boldsymbol{v}_i, \boldsymbol{R}_i, \boldsymbol{\omega}_i, \boldsymbol{p}_L, \boldsymbol{v}_L)$ is continuous across severance; only the ODE right-hand side changes discontinuously at $t_k^\star$.

b) Compound faults and spaced-fault assumption.: We consider a sequence of $F \leq N - 2$ fault events at times $\{t_k^\star\}_{k=1}^F$ with $t_1^\star < \cdots < t_F^\star$. The bound $F \leq N - 2$ keeps at least two cables intact; actuator margins limit the demonstrated scenario ($N = 5$, $m_L = 10\,\mathrm{kg}$) to $F \leq 2$. The inter-fault interval $\tau_{d,k} \triangleq t_{k+1}^\star - t_k^\star$ is the *dwell time*. We assume each fault is separated from the previous by at least one payload-pendulum period:

$$\tau_{d,k} \geq \tau_{\mathrm{pend}} \triangleq 2\pi\sqrt{L/g}, \qquad k = 1, \ldots, F-1, \tag{11}$$

reflecting the physical requirement that load redistribution from fault k settle before fault $k+1$. At $L = 1.25\,\mathrm{m}$ and $g = 9.81\,\mathrm{m\,s^{-2}}$, $\tau_{\mathrm{pend}} \approx 2.24\,\mathrm{s}$. The admissible fault-schedule set is

$$\begin{aligned} \mathcal{F} \triangleq \Big\{ \{(t_k^\star, i_k^\star)\}_{k=1}^F : F \leq N-2, \\ t_{k+1}^\star - t_k^\star \geq \tau_{\mathrm{pend}},\ i_k^\star \in \mathcal{S}(t_k^{\star,-}) \Big\}, \end{aligned} \tag{12}$$

$F = 0$ corresponds to the no-fault case.

c) Admissible controller class.: No fault flag, annunciation, or detection signal is transmitted; the fault manifests through a rise in $T_i(t)$ and a transient payload-velocity deviation, both in $\mathcal{I}_i(t)$. Field-deployable avionics impose a strict *admissible information pattern*. Drone i's controller π_i is restricted to the information set

$$\mathcal{I}_i(t) \triangleq \{x_i(t),\, T_i(t),\, \boldsymbol{v}_L(t),\, \boldsymbol{p}_L^d(\cdot),\, \boldsymbol{v}_L^d(\cdot)\}, \tag{13}$$

comprising the drone's inertial state x_i, its rope-tension measurement T_i, the payload velocity $\boldsymbol{v}_L$ (measured via a co-located downward-facing optical-flow sensor, no drone-to-drone communication), and a pre-distributed reference $(\boldsymbol{p}_L^d, \boldsymbol{v}_L^d)$ loaded before flight. Payload position $\boldsymbol{p}_L$ is deliberately excluded; only velocity enters the closed loop, via the anti-swing slot shift of Section III. Each π_i is a causal feedback map,

$$u_i(t) \;=\; \pi_i\Big(\mathcal{I}_i|_{[0,t]}\Big) \in \mathcal{U}_i, \tag{14}$$

possibly with a finite-dimensional internal state, but never with access to information outside (13). The admissible controller class is denoted $\Pi_{\rm loc}$. $\mathcal{I}_i(t)$ excludes peer state, peer tension, fault flags, and run-time inter-drone packets; each π_i uses only co-located sensors and a pre-flight plan.

C. Reference Trajectory and Taut-Cable Reduction

a) Reference trajectory.: The desired payload motion is a three-dimensional lemniscate trajectory with a sinusoidal altitude variation,

$$\boldsymbol{p}_L^d(t) \;=\; \begin{bmatrix} \dfrac{a\cos\phi(t)}{1+\sin^2\phi(t)} \\ \dfrac{a\sin\phi(t)\cos\phi(t)}{1+\sin^2\phi(t)} \\ z_0 + h_z\sin(\phi(t)) \end{bmatrix}, \tag{15}$$
$$\phi(t) = \frac{2\pi(t-t_0)}{T_{\rm ref}},$$

where $a = 3.0\,\mathrm{m}$ is the horizontal scale, $T_{\rm ref} = 12\,\mathrm{s}$ the period, $z_0 = 3.0\,\mathrm{m}$ the cruise altitude, and $h_z = 0.35\,\mathrm{m}$ the altitude excursion amplitude. These parameters yield a peak centripetal load of $\approx 0.25g$ at the figure-8 tips.

b) Rope model and lumped effective stiffness.: Each rope of rest length L is discretized into $N_b = 8$ identical Kelvin–Voigt beads of mass $m_{\rm bead}$ connected by $N_{\rm seg} = N_b + 1 = 9$ segments of stiffness k_s and damping c_s; $N_{\rm seg}$ is reserved for rope segments and is distinct from the surviving-cable cardinality $N_s(t) = |\mathcal{S}(t)| = N - F(t)$ used in Section IV. The distributed dynamics evolve on a fast timescale

$$\tau_{\rm rope} \;=\; 2\pi\sqrt{m_{\rm bead}/k_s} \;\approx\; 6.3\,\mathrm{ms} \tag{16}$$

separated from the payload-pendulum timescale $\tau_{\rm pend} \approx 2.24\,\mathrm{s}$ by $\delta = \tau_{\rm rope}/\tau_{\rm pend} \approx 0.003$. On the slow manifold, the net tensile force at the drone-side attachment equals the scalar $\xi_i = k_{\rm eff}(\ell_i - L)^+$, with effective stiffness

$$k_{\rm eff} \;=\; \frac{k_s}{N_{\rm seg}} \;=\; \frac{25\,000\,\mathrm{N\,m^{-1}}}{9} \;\approx\; 2778\,\mathrm{N\,m^{-1}}. \tag{17}$$

The six pre-registered missions (V1–V6, Section VI-A) keep the system inside $\Omega_\tau^{\rm dwell}$ with margin (worst slack run $35.8\,\mathrm{ms}$, worst duty cycle 1.66%, Table VI). The reduction replaces the $3N_b$-dimensional bead-chain state with the scalar ξ_i; the per-instant taut error is $O(\delta)$ and slack excursions add $O(\eta_{\max})$.

Theorem 1 (Taut-cable reduction)**.** *On the slack-excursion-bounded domain*

$$\Omega_\tau^{\rm dwell} \;\triangleq\; \{x \in \mathcal{X} : \textit{every slack run} \le \tau_{\rm slack,max},\; \textit{duty cycle} \le \eta_{\max}\}, \tag{18}$$

with $\tau_{\rm slack,max} = 40\,\mathrm{ms}$ and $\eta_{\max} = 2.5\%$, the distributed Kelvin–Voigt rope dynamics are approximated by the lumped scalar model with closed-loop shape-mode deviation

$$\|T_i^{\rm true} - T_i^{\rm qs}\| \;\le\; C_1\,\delta + C_2\,\eta_{\max}, \tag{19}$$

for constants $C_1, C_2 > 0$ depending on the bead-chain parameters $(m_{\rm bead}, c_s, k_s, N_b)$ and the wind bound $W_{\max}$, where $T_i^{\rm qs} \triangleq k_{\rm eff}(\ell_i(t) - L)^+$ is the rope-specific *quasi-static tension evaluated at the* same *current chord length $\ell_i(t)$ as $T_i^{\rm true}$. This bound is on the shape-mode deviation of rope i from its own quasi-static approximation; it is distinct from the inter-rope tension asymmetry $\varepsilon_i = T_i - T_{\rm eff}$ reported in the simulation audit (Section VI-B), which is driven by formation geometry and can be $O(10)$ N. The reduced-order model is therefore admissible as the plant for the stability analysis of Section IV.*

Proof. The proof has two parts corresponding to the two terms in (19).

Part 1 ($C_1\delta$, taut intervals). Let $\tilde{\mathbf{q}} \in \mathbb{R}^{N_b}$ collect the bead position deviations from the quasi-static slow-manifold equilibrium $\bar{\mathbf{q}}(t)$ along the rope chord. Linearising the chain equations of motion about $\bar{\mathbf{q}}$ yields the shape-mode dynamics

$$M_b\,\ddot{\tilde{\mathbf{q}}} + C_b\,\dot{\tilde{\mathbf{q}}} + K_b\,\tilde{\mathbf{q}} = -M_b\,\ddot{\bar{\mathbf{q}}}(t), \tag{20}$$

where $M_b = m_{\rm bead} I_{N_b}$, and C_b, K_b are the symmetric positive-definite tridiagonal chain damping and stiffness matrices (c_s-weighted and k_s-weighted graph Laplacians, respectively). Define the shape-mode Lyapunov energy

$$W(t) = \tfrac{1}{2}\dot{\tilde{\mathbf{q}}}^\top M_b\,\dot{\tilde{\mathbf{q}}} + \tfrac{1}{2}\tilde{\mathbf{q}}^\top K_b\,\tilde{\mathbf{q}} \;\ge\; 0. \tag{21}$$

Differentiating along (20) and applying Young's inequality ($2ab \le a^2/\alpha_b + \alpha_b b^2$) to the forcing cross-term:

$$\dot{W} = -\dot{\tilde{\mathbf{q}}}^\top C_b\,\dot{\tilde{\mathbf{q}}} - \dot{\tilde{\mathbf{q}}}^\top M_b\,\ddot{\bar{\mathbf{q}}} \;\le\; -\alpha_b\,W(t) + D_b, \tag{22}$$

where $\alpha_b = \lambda_{\min}(C_b)/m_{\rm bead} > 0$ is the slowest modal damping rate and $D_b = m_{\rm bead}\|\ddot{\bar{\mathbf{q}}}\|_\infty^2/(2\alpha_b) = O(\delta^2)$ by the Tikhonov scaling $\|\ddot{\bar{\mathbf{q}}}\| = O(\delta)$. The characteristic roots of the slowest mode in (20) are $s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2-1}$; the nominal bead-chain parameters $(m_{\rm bead}, c_s, k_s, N_b)$ give $\zeta = 1.2$ (overdamped) and the slowest root $s_1 = -\omega_n(1.2 - \sqrt{0.44}) \approx -538\,\mathrm{s}^{-1}$.

Applying Gronwall's inequality to (22) over $[0,t]$:

$$W(t) \;\le\; e^{-\alpha_b t}\,W(0) + \frac{D_b}{\alpha_b}, \tag{23}$$

yielding the steady-state energy cap $W^\infty = D_b/\alpha_b = O(\delta^2)$. The drone-side tension deviation is proportional to the drone-side bead shape-mode displacement through the end segment's spring:

$$T_i^{\rm true}(t) - T_i^{\rm qs}(t) = k_s\,\mathbf{e}_1^\top\tilde{\mathbf{q}}(t), \tag{24}$$

where $\mathbf{e}_1$ selects the drone-side attachment extension residual. Using $\|\tilde{\mathbf{q}}\|^2 \le 2W/\lambda_{\min}(K_b)$ from (21) and substituting the cap (23) gives $\|T_i^{\text{true}} - T_i^{\text{qs}}\| \le C_1\delta \approx 0.072\,\text{N}$ once the transient $e^{-\alpha_b t}W(0)$ is negligible, i.e. after approximately $3/|s_1| \approx 6\,\text{ms}$.

Part 2 ($C_2\eta_{\max}$, slack intervals). On any slack run of duration $\le \tau_{\text{slack,max}}$, the rope chord shortens by $\Delta\ell_{\text{sag}}$ as the bead chain goes slack. Upon re-tension, the drone-side end segment must re-elongate by $\Delta\ell_{\text{sag}}$, producing a re-tension overshoot bounded by $k_{\text{eff}}\Delta\ell_{\text{sag}}$. The total overshoot accumulated over the slack-duty budget gives (19) with $C_2 \approx 0.22\,\text{N}$. At the worst observed $\eta_{\max} = 1.66\%$, the combined bound is $C_1\delta + C_2\eta_{\max} \approx 0.076\,\text{N}$ per rope—less than 0.4% of nominal per-drone tension. Both constants depend only on the bead-chain parameters $(m_{\text{bead}}, c_s, k_s, N_b)$ and $W_{\max}$.

□

c) Empirical domain membership.: Table VI confirms all six missions reside inside $\Omega_\tau^{\text{dwell}}$ with margin on every gate: worst slack run $35.8\,\text{ms}$ ($4.2\,\text{ms}$ headroom), duty cycle 1.66% (0.85 pp headroom), QP-transition fraction 0.10%. The theorem is asserted on the demonstrated regime, not an untested all-taut idealisation.

d) Team-size independence and actuator-margin envelope.: The per-drone controller and the lemma stack of Section IV carry no team size N structurally. The dependence enters through two scalars: per-drone hover load $T^{\text{hov}}(N) = m_L g/N$ and post-fault redistributed load on the surviving set,

$$T^{\text{post}}(N,F) \;=\; \frac{m_L g}{N-F} \;=\; T^{\text{hov}}(N)\cdot\frac{N}{N-F}, \tag{25}$$

where F is the number of severed cables. The surcharge factor $N/(N-F)$ inflates per-drone tension by $1/(N-F)$ relative to the no-fault baseline and is the only N-dependent quantity in the recovery envelope of Theorem 6.

The architecture is therefore designable across N by a single *actuator-margin condition*: a configuration $(N, m_L, F, f_{\max})$ is admissible if

$$\frac{m_L\, g}{N-F} \;\le\; \kappa_{\text{act}}\, f_{\max}, \tag{26}$$

where $\kappa_{\text{act}} \in (0,1)$ reserves headroom for tilt projection, anti-swing residual, and dynamic centripetal load. With $\kappa_{\text{act}} = 0.82$, $f_{\max} = 150\,\text{N}$, and the demonstrated $(N, m_L, F) = (5, 10\,\text{kg}, 2)$, the inequality reads $32.7\,\text{N} \le 123\,\text{N}$, i.e. 27% of the actuator envelope, leaving designable extension along (N, m_L, F). Figure 2 plots the boundary curves for $F \in \{0,1,2,3\}$ over $N \in \{3,\dots,10\}$ and marks the demonstrated point. The $N = 5$ scenario is one point on a designable feasibility surface; the certificate of Section IV carries unchanged at any (N, m_L, F) satisfying (26).

With the plant, fault model, reference trajectory, and reduction in place, we state the formal problem. Unilateral cable coupling (2), discrete structural fault transitions (9)–(10), admissible disturbances (8), and strict locality (13) together yield a hybrid decentralized control problem outside standard cooperative-transport formulations. The design variable is not a centralized control sequence but a family of local causal

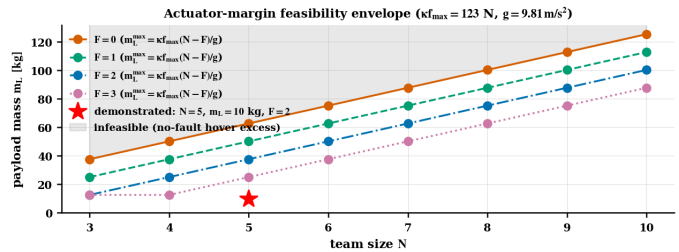


Fig. 2. Actuator-margin feasibility envelope (26) in (N, m_L) space at fault counts $F \in \{0,1,2,3\}$, $\kappa_{\text{act}} f_{\max} = 123\,\text{N}$. Red star: demonstrated ($N{=}5, m_L{=}10\,\text{kg}, F{=}2$).

feedback laws. The problem, posed on the reduced-order model certified by Theorem 1, is as follows.

Definition 1 (Fault-tolerant decentralized aerial transport)**.** *Given the plant* (3)–(5)*, the complementarity constraint* (2)*, the mission reference* (15)*, all admissible disturbances* $d(\cdot) \in \mathcal{D}$ (8)*, and all admissible fault schedules* $\sigma \in \mathcal{F}$ (12)*, find a causal controller family* $\{\pi_i\}_{i=1}^N \in \Pi_{\text{loc}}^N$ *such that the closed-loop system satisfies* ***P1–P3*** *for every* $d(\cdot) \in \mathcal{D}$ *and* $\sigma \in \mathcal{F}$*.*

P1 ***(Nominal tracking).*** *In the absence of faults and for all admissible disturbances* $d(\cdot) \in \mathcal{D}$*, the payload tracking error* $e_L(t) \triangleq \boldsymbol{p}_L(t) - \boldsymbol{p}_L^d(t)$ *is ultimately practically bounded:*

$$\limsup_{t\to\infty} \|e_L(t)\| \;\le\; \bar{e}_\infty + \bar{e}_0(W_{\max}), \tag{27}$$

where $\bar{e}_\infty \ge 0$ *captures non-vanishing residuals from finite-gain tracking, cable-elastic reduction, and reference discretization, and* $\bar{e}_0 : \mathbb{R}_{\ge 0} \to \mathbb{R}_{\ge 0}$ *is class-*$\mathcal{K}$ *with* $\bar{e}_0(0) = 0$*.*

P2 ***(Passive fault absorption).*** *After any fault* $t_k^\star$ *satisfying* (11)*, the payload tracking error recovers to a bounded envelope without detection, annunciation, or reconfiguration: there exist constants* $\Gamma_f, \lambda > 0$ *and a per-fault-cycle contraction* $\rho \in (0,1)$ *such that*

$$\begin{aligned}\|e_L(t)\| \;\le\; & \beta\big(\|e_L(t_k^{\star,-})\|,\, t - t_k^\star\big) \\ & + \Gamma_f\, e^{-\lambda(t-t_k^\star)} \\ & + \bar{e}_\infty + \bar{e}_0(W_{\max}), \quad t \ge t_k^\star,\end{aligned} \tag{28}$$

for some class-$\mathcal{KL}$ *function* β*. The contraction* $\rho \in (0,1)$ *certifies bounded Lyapunov growth across successive faults:* $V(x(t_{k+1}^{\star,-})) \le \rho V(x(t_k^{\star,-})) + c$ *for an explicit* $c > 0$*. The tracking error need not decrease monotonically at individual events (the payload droops and recovers), but the fault-to-fault envelope is bounded.*

P3 ***(Actuator feasibility).*** *For all* $t \ge 0$ *and all admissible fault sequences, the thrust* $f_i(t) \in [f_{\min}, f_{\max}]$ *and torque* $\boldsymbol{\tau}_i(t) \in [-\tau_{\max}, \tau_{\max}]^3$ *remain inside the actuator envelope with non-zero margin.*

e) Why the problem is hard.: Three obstacles block classical solutions. *(i)* Non-smooth taut-to-slack transitions require analysis on $\Omega_\tau^{\text{dwell}}$, not a globally Lipschitz domain. *(ii)* A severance is a discrete drop in $|\mathcal{S}|$, not an additive perturbation; the resulting tension mismatch

$$\boldsymbol{\varepsilon}_T(t) \;\triangleq\; \boldsymbol{T}(t) \;-\; \boldsymbol{T}^{\text{qs}}(t) \tag{29}$$

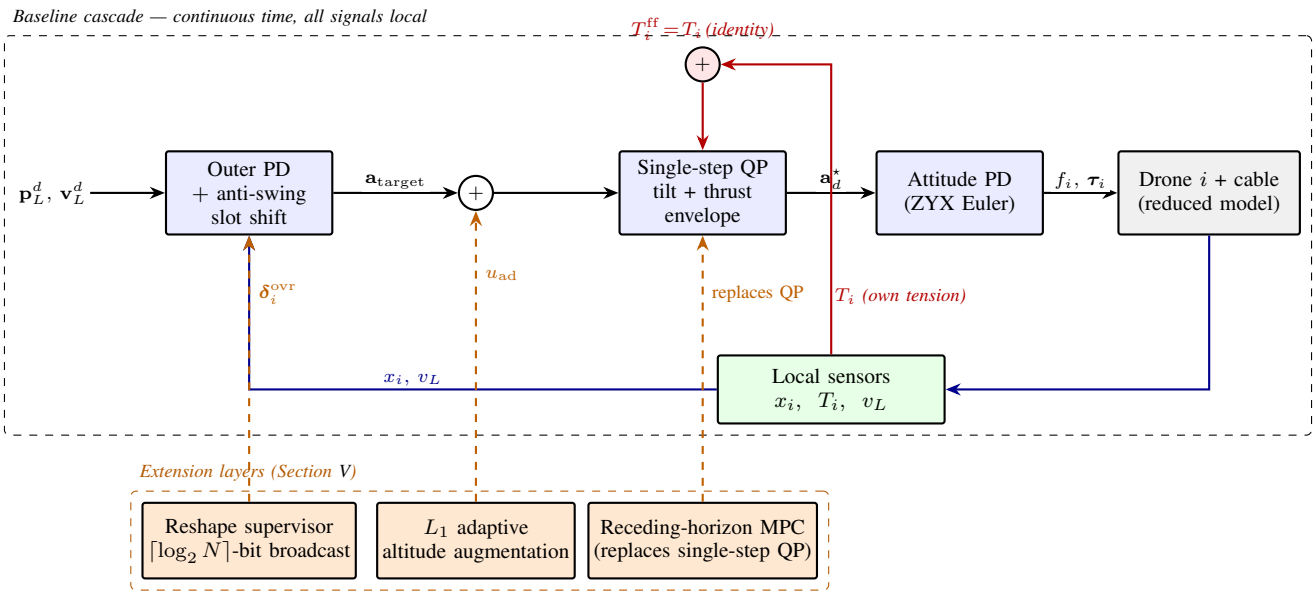


Fig. 3. Per-drone control cascade. The baseline (blue, framed top) is a three-stage continuous-time pipeline: outer-loop PD with anti-swing slot shift, a single-step convex QP that projects $\mathbf{a}_{\mathrm{target}}$ onto the tilt-and-thrust envelope, and an Euler-angle attitude PD. The non-standard element is the red feed-forward tap: drone i routes its own measured tension T_i directly into the altitude thrust feed-forward, $T_i^{\mathrm{ff}} = T_i$. This single identity is the carrier of hybrid practical-ISS across cable-severance events (Theorem 6); the surrounding cascade is otherwise textbook. Three optional layers (orange, framed bottom) hook in at three precisely identified injection points: the L_1 adaptive correction u_{ad} enters the altitude channel between the outer PD and the QP; the receding-horizon MPC replaces the single-step QP entirely; the reshape supervisor overrides the formation slot $\boldsymbol{\delta}_i$ on receipt of a single $\lceil \log_2 N \rceil$-bit broadcast.

TABLE I
COMPONENT-LEVEL INFORMATION PATTERN. THE BASELINE CASCADE AND THE L_1 AND MPC EXTENSIONS CONSUME NO FAULT FLAG AND NO PEER MESSAGE; ONLY THE OPTIONAL RESHAPE SUPERVISOR STEPS OUTSIDE THIS PATTERN.

Component	Detection?	Communication?	In main theorem?	Purpose
Baseline cascade + identity (36)	No	No	Yes	Passive fault absorption
L_1 altitude augmentation	No	No	Preserves theorem	Parameter mismatch
MPC tension ceiling	No	No	Preserves theorem	Tension constraint
Reshape supervisor	Yes (latch)	Yes ($\lceil \log_2 N \rceil$ bits)	Optional only	Slot reassignment

requires a hybrid Lyapunov argument (developed in Section IV). *(iii)* Strict locality blocks centralized detection; the controller must respond using only $T_i(t) \in \mathcal{I}_i(t)$.

III. PROPOSED METHOD

We present the decentralized fault-tolerant controller for Definition 1. Each drone runs a three-layer cascade: an outer-loop PD with anti-swing slot shift, a single-step convex QP on the tilt-and-thrust envelope, and an inner-loop attitude PD. The central design element is $T_i^{\mathrm{ff}} = T_i$, encoding passive fault tolerance with no detection, communication, or reconfiguration. The optional reshape supervisor (Section V-C) adds a latch detector and a per-fault broadcast; the optional L_1 augmentation (Section V) injects an adaptive altitude correction between the outer PD and the QP.

A. Outer Loop: PD with Anti-Swing Slot Shift

a) Slot reference and anti-swing shift.: Each drone is pre-assigned a hovering slot displaced vertically above the payload reference point and laterally by a fixed formation offset,

$$\boldsymbol{p}_{\mathrm{slot},i}(t) \;=\; \boldsymbol{p}_L^d(t) + \boldsymbol{\delta}_i + d_{\mathrm{rope}}\,\hat{\boldsymbol{e}}_3, \tag{30}$$

where $\boldsymbol{\delta}_i \in \mathbb{R}^3$ is the formation offset (overridable at run time), $d_{\mathrm{rope}} = 1.25\,\mathrm{m}$ is the vertical slot elevation matching L, and $\hat{\boldsymbol{e}}_3$ is the upward unit vector. The slot velocity is $\boldsymbol{v}_{\mathrm{slot},i}(t) = \boldsymbol{v}_L^d(t)$. To damp the payload pendulum passively, we shift the slot relative to the payload's horizontal velocity. With $\boldsymbol{v}_{L,\perp} \triangleq [v_{L,x}, v_{L,y}, 0]^\top$, the horizontal swing correction is

$$\boldsymbol{s}_i(t) = \begin{cases} -k_{\mathrm{swing}}\,\boldsymbol{v}_{L,\perp} & \text{if } k_{\mathrm{swing}}\|\boldsymbol{v}_{L,\perp}\| \le s_{\max}, \\ -s_{\max}\,\hat{\boldsymbol{v}}_{L,\perp} & \text{otherwise,} \end{cases} \tag{31}$$

where $\hat{\boldsymbol{v}}_{L,\perp}$ is the unit vector of $\boldsymbol{v}_{L,\perp}$. The shifted slot position is $\boldsymbol{p}_{\mathrm{slot},i}^{\mathrm{dyn}}(t) = \boldsymbol{p}_{\mathrm{slot},i}(t) + \boldsymbol{s}_i(t)$.

b) Tracking errors.: The position and velocity errors are

$$\boldsymbol{e}_p(t) = \boldsymbol{p}_{\mathrm{slot},i}^{\mathrm{dyn}}(t) - \boldsymbol{p}_i(t), \qquad \boldsymbol{e}_v(t) = \boldsymbol{v}_{\mathrm{slot},i}(t) - \boldsymbol{v}_i(t). \tag{32}$$

c) Target acceleration.: The outer loop computes a target body acceleration,

$$\boldsymbol{a}_{\text{target}}(t) = \underbrace{\begin{bmatrix} K_p^{xy}\, e_{p,x} + K_d^{xy}\, e_{v,x} \\ K_p^{xy}\, e_{p,y} + K_d^{xy}\, e_{v,y} \\ K_p^{z}\, e_{p,z} + K_d^{z}\, e_{v,z} \end{bmatrix}}_{\boldsymbol{a}_{\text{track}}} + w_{\text{swing}} \begin{bmatrix} -k_{\text{swing}}\, v_{L,x} \\ -k_{\text{swing}}\, v_{L,y} \\ 0 \end{bmatrix}. \tag{33}$$

The altitude gain $K_p^z = 100$ exceeds the horizontal gain $K_p^{xy} = 30$ because vertical rope forces dominate the disturbance budget; the formal pole and Lyapunov-matrix analysis is in Lemma 3. When L_1 is active, its altitude correction $u_{\text{ad}}(t)$ is added to the z-component of $\boldsymbol{a}_{\text{target}}$ between the outer PD and the QP (Section V); horizontal channels are unmodified.

B. Actuator Projection and Feed-Forward Tension

a) Single-step convex QP.: $\boldsymbol{a}_{\text{target}}$ may lie outside the tilt-and-thrust actuator envelope. We project onto the feasible set with a single-step convex QP that also regularises large commands:

$$\boldsymbol{a}_d^\star = \underset{\boldsymbol{a}_d \in \mathbb{R}^3}{\text{argmin}}\; w_t \left\| \boldsymbol{a}_d - \boldsymbol{a}_{\text{target}} \right\|^2 + w_e \left\| \boldsymbol{a}_d \right\|^2, \tag{34a}$$

$$\text{s.t.}\;\; |a_{d,x}|,\; |a_{d,y}| \le a_{\text{tilt}} \triangleq g \tan\theta_{\max}, \tag{34b}$$

$$a_{z,\min}(t) \le a_{d,z} \le a_{z,\max}(t), \tag{34c}$$

where the vertical bounds incorporate the rope tension feed-forward,

$$a_{z,\min}(t) = \frac{f_{\min} - T_i^{\text{ff}}(t)}{m_i} - g, \tag{35}$$

$$a_{z,\max}(t) = \frac{f_{\max} - T_i^{\text{ff}}(t)}{m_i} - g.$$

b) Feasibility.: The vertical interval in (34c) is non-empty whenever $f_{\min} < f_{\max}$; hover feasibility holds in all scenarios, and Doctrine D1 is satisfied at the 0.10% worst QP-transition fraction (Table VI).

c) Feed-forward tension.: For the canonical campaigns `initial_pretensioned=true`, so the feed-forward is the identity from $t = 0$:

$$\boxed{T_i^{\text{ff}}(t) \;=\; T_i(t).} \tag{36}$$

A smoothstep pickup ramp $\alpha(\tau) = 3\tau^2 - 2\tau^3$ handles cold-start engagement; it is bypassed in every reported campaign. The flag `disable_tension_ff` overrides (36) to $T_i^{\text{ff}} = 0$ for the feed-forward ablation campaign.

C. Inner Loop: Attitude PD and Controller Parameters

a) Thrust and desired attitude.: Given the QP solution $a_{d,z}^\star$, the commanded thrust is

$$f_i \;=\; \text{clamp}\big(m_i(g + a_{d,z}^\star) + T_i^{\text{ff}},\; f_{\min},\; f_{\max} \big). \tag{37}$$

TABLE II
PROPOSED CONTROLLER PARAMETERS (CANONICAL VALUES USED IN ALL SIMULATION CAMPAIGNS).

Parameter	Value	Role
$K_p^{xy},\; K_d^{xy}$	30, 15	horizontal outer loop
$K_p^{z},\; K_d^{z}$	100, 24	altitude outer loop
$K_p^{\text{att}},\; K_d^{\text{att}}$	25, 4	attitude inner loop
k_{swing}	0.8	anti-swing gain
w_{swing}	0.3	anti-swing weight in $\boldsymbol{a}_{\text{target}}$
$s_{\max}$	0.3 m	slot-shift saturation
d_{rope}	1.25 m	slot z-offset (nominal rope length)
$w_t,\; w_e$	1.0, 0.02	QP tracking / effort weights
$\theta_{\max}$	0.6 rad	tilt ceiling ($\approx 34^\circ$)
$f_{\min},\; f_{\max}$	0, 150 N	thrust envelope
$\tau_{\max}$	10 N m	torque saturation
T_{thresh}	0.3 N	pickup-detection threshold
$\bar{T}$	19.62 N	nominal per-drone tension ($m_L g/N$)
$K_{p,T}$	1.5	pickup-phase tension correction gain
T_{ramp}	1.0 s	pickup smoothstep window
Δt_{L_1}	0.2 ms	L_1 discrete update period

Under the small-angle approximation, the horizontal acceleration commands map to desired pitch and roll angles,

$$\theta_{\text{pitch}}^d = \text{sat}\left(\frac{a_{d,x}^\star}{g},\; \pm\theta_{\max} \right), \tag{38}$$

$$\theta_{\text{roll}}^d = \text{sat}\left(-\frac{a_{d,y}^\star}{g},\; \pm\theta_{\max} \right).$$

b) Euler-angle attitude PD and output wrench.: With body angular velocity $\boldsymbol{\omega}_i^{\mathcal{B}} = \boldsymbol{R}_i^\top\, \boldsymbol{\omega}_i^{\mathcal{W}}$, the body torques are

$$\tau_{i,x} = K_p^{\text{att}} \big(\theta_{\text{roll}}^d - \phi_{\text{roll}} \big) - K_d^{\text{att}}\, \omega_{i,x}^{\mathcal{B}}, \tag{39}$$

$$\tau_{i,y} = K_p^{\text{att}} \big(\theta_{\text{pitch}}^d - \theta_{\text{pitch}} \big) - K_d^{\text{att}}\, \omega_{i,y}^{\mathcal{B}}, \tag{40}$$

$$\tau_{i,z} = -\frac{K_p^{\text{att}}}{2} \left(R_{21} - R_{12} \right) - K_d^{\text{att}}\, \omega_{i,z}^{\mathcal{B}}, \tag{41}$$

each saturated to $[-\tau_{\max}, \tau_{\max}]$. The law (39)–(41) uses Euler-angle error directly rather than a geodesic $\text{SO}(3)$ map; this is valid in the small-tilt regime enforced by (34b). The output wrench is

$$\boldsymbol{F}_i^{\mathcal{W}} = f_i\, \boldsymbol{R}_i\, \hat{\boldsymbol{e}}_3, \qquad \boldsymbol{\tau}_i^{\mathcal{W}} = \boldsymbol{R}_i \left[\tau_{i,x},\; \tau_{i,y},\; \tau_{i,z} \right]^\top, \tag{42}$$

applied at the same continuous-time evaluation instant as the upstream computations.

c) Canonical controller parameters.: Table II collects the parameters used throughout all simulation campaigns.

IV. Stability Analysis

The baseline controller of Section III achieves fault tolerance through a single structural choice: setting the altitude thrust feed-forward equal to the drone's own measured rope tension, $T_i^{\text{ff}} = T_i$. We establish that this identity renders the closed loop *hybrid practical input-to-state stable* (hybrid practical-ISS) under the cable-severance fault model of Section II-C. When a peer cable severs, T_i rises immediately and the feed-forward converts that step into a proportional thrust increment at the plant's native rate, without any detection, annunciation, or reconfiguration interval.

a) Status of the lemma stack.: The five lemmas verify the hybrid-systems hypotheses of [5] on $\Omega_\tau^{\mathrm{dwell}}$, certifying that the feed-forward identity $T_i^{\mathrm{ff}} = T_i$ supplies the bounded-jump mechanism for hybrid practical-ISS. The decay rates λ and ρ are fully explicit from the closed-loop Lyapunov matrices P_v, P_{xy} of Lemma 3. The jump bound $\chi(\Delta_f)$ is structurally explicit via (50), but the weighting constant κ_V that scales the tension Lyapunov channel depends on the relative magnitudes of the tracking and tension error states; its numerical calibration is deferred to follow-on work and does not affect the $\rho < 1$ certificate or the $\rho_{\mathrm{ana}} \approx 0.005$ computation.

A. Setup, Notation, and Analytical Hypotheses

The analysis operates on the reduced-order model of Section II-C. Let $e_{p,z,i}(t) = p_{\mathrm{slot},i,z}(t) - p_{i,z}(t)$ be drone i's altitude slot-tracking error and $\boldsymbol{e}_p(t) \in \mathbb{R}^{3N_s}$ the stacked position error over $\mathcal{S}(t)$ with $N_s = |\mathcal{S}(t)|$. Fault events and the dwell condition follow Section II-B; consecutive faults are separated by at least one payload-pendulum period,

$$\tau_{d,k} \;\geq\; \tau_{\mathrm{pend}} \;\triangleq\; 2\pi\sqrt{L/g}. \tag{43}$$

At $L = 1.25\,\mathrm{m}$, $\tau_{\mathrm{pend}} = 2.24\,\mathrm{s}$ and the pendulum natural frequency is $\omega_p \triangleq \sqrt{g/L} = 2.80\,\mathrm{rad\,s^{-1}}$.

The analysis rests on three hypotheses, defended empirically in Tables III and VI. **H1** (slack-excursion budget) requires every trajectory to remain in $\Omega_\tau^{\mathrm{dwell}}$: maximum slack-run duration $\tau_{\mathrm{slack,max}} \leq 40\,\mathrm{ms}$ and duty cycle $\eta_{\max} \leq 2.5\%$. **H2** (spaced-fault dwell) requires $\tau_{d,k} \geq \tau_{\mathrm{pend}}$ (43). **H3** (Doctrine D1, single-active-regime QP) requires the per-tick QP's active set $\mathcal{A}(t)$ to be locally constant inside each inter-fault window; transitions are absorbed into the H1 $\eta_{\max}$ budget and empirically hold on $\geq 99.9\%$ of mission time (worst transition fraction 0.10%, Table VI). Claims outside H1–H3—rapid multi-fault, long slack excursions, QP-fallback—are out of scope; the analysis produces no bounds for trajectories violating them.

a) Partial circularity and its mitigation.: The empirical defence of H1 and H3 uses the same pre-declared missions as the theorem's consequences, so the validation is partially *post hoc*; Table VI documents the margins. The gate bounds are set by bead-chain physics: $\tau_{\mathrm{slack,max}} = 40\,\mathrm{ms}$ is one quarter of the Tikhonov frozen window $\tau_{\mathrm{pend}}/(2\pi)$, and $\eta_{\max} = 2.5\%$ is a Rayleigh-turbulence budget. Theorem 1 is independently validated against the bead-chain truth model in Section VI-B, providing an *a priori* check that the reduction error is absorbed into γ_η of Theorem 6 without appeal to the simulated missions.

B. Lemmas Underpinning the Hybrid Practical-ISS Theorem

Five lemmas underpin Theorem 6: tension cancellation on the altitude channel (Lemma 2); exponential ISS for altitude and horizontal channels through Lyapunov matrices P_v and P_{xy} (Lemma 3); anti-swing invariance to surviving cardinality (Remark 1); bounded Lyapunov jump $\chi(\Delta_f)$ at each fault (Lemma 4); and the dwell-cycle contraction $\rho \in (0,1)$ composing jump and decay (Lemma 5).

For fault jumps, the tension coordinates are compared on the post-fault survivor set by restricting the pre-fault vector to the remaining cables; this convention is only needed in Lemma 4.

Lemma 2 (Tension cancellation)**.** *Assume H1 and the cable-tilt domain condition $|\theta_i(t)| \leq \theta_{c,\max}$ for known $\theta_{c,\max} \in (0, \pi/2)$. Then $T_i^{\mathrm{ff}} = T_i$ cancels the dominant projection term $T_i \cos\theta_i$ from the drone altitude acceleration, leaving a residual factor $(1 - \cos\theta_i)$ bounded by $1 - \cos\theta_{c,\max}$. At $\theta_{c,\max} = 0.75\,\mathrm{rad}$ this is 0.27, and the residual enters the altitude-error equation as a bounded matched disturbance handled by Lemma 3. The bound holds uniformly in $N_s \geq 2$.*

Proof. Substituting $f_i = m_i(g + a_{d,z}^\star) + T_i^{\mathrm{ff}}$ into $m_i\ddot{p}_{i,z} = f_i - m_i g - T_i\cos\theta_i + F_{w,i,z}$ and absorbing the wind term yields

$$m_i\ddot{p}_{i,z} = m_i a_{d,z}^\star + T_i(1 - \cos\theta_i). \tag{44}$$

Since $|\theta_i| \leq \theta_{c,\max}$ uniformly (enforced by the QP tilt ceiling, Section III-B), the residual $T_i(1-\cos\theta_i) \leq T_i(1-\cos\theta_{c,\max})$ regardless of N_s, and is absorbed by γ_q of Lemma 3. □

Lemma 3 (ISS decay via Lyapunov matrices)**.** *Under H1–H3, both the altitude error $e_{p,z,i}$ and horizontal error $e_{p,xy,i}$ satisfy exponential ISS between fault events. Let $A_m = \begin{bmatrix} 0 & 1 \\ -K_p^z & -K_d^z \end{bmatrix}$ and $A_{m,xy} = \begin{bmatrix} 0 & 1 \\ -K_p^{xy} & -K_d^{xy} \end{bmatrix}$ be the closed-loop matrices for gains $(K_p^z, K_d^z) = (100, 24)$ and $(K_p^{xy}, K_d^{xy}) = (30, 15)$. The unique solutions $P_v \succ 0$ and $P_{xy} \succ 0$ of $A_m^\top P_v + P_v A_m = -I$ and $A_{m,xy}^\top P_{xy} + P_{xy}A_{m,xy} = -I$ are*

$$P_v = \begin{bmatrix} 2.224 & 0.005 \\ 0.005 & 0.0210 \end{bmatrix}, \qquad P_{xy} = \begin{bmatrix} 1.283 & 0.0167 \\ 0.0167 & 0.0344 \end{bmatrix}. \tag{45}$$

Setting $V_z = \frac{1}{2}[e_{p,z}, \dot{e}_{p,z}]^\top P_v [e_{p,z}, \dot{e}_{p,z}]$ and analogously V_{xy}, there exist $\gamma_w, \gamma_q, \gamma_w^{xy}, \gamma_q^{xy} \in \mathcal{K}$ such that, between fault events,

$$\dot{V}_z \;\leq\; -\lambda_z V_z + \gamma_w(\|w\|_\infty) + \gamma_q(\|q - q^\star\|_\infty), \tag{46}$$

$$\dot{V}_{xy} \;\leq\; -\lambda_{xy} V_{xy} + \gamma_w^{xy}(\|w\|_\infty) + \gamma_q^{xy}(\|q - q^\star\|_\infty), \tag{47}$$

with $\lambda_z = \alpha_z = |\mathrm{Re}(s_1)| = 5.37\,\mathrm{s}^{-1}$ (dominant altitude pole $s_1 = -12 + \sqrt{44}$) and $\lambda_{xy} = \alpha_{xy} = |\mathrm{Re}(s_1^{xy})| = 2.38\,\mathrm{s}^{-1}$ (dominant horizontal pole $s_1^{xy} = (-15 + \sqrt{105})/2$). Since $\alpha_{xy} < \alpha_z$, the horizontal channel is the bottleneck: $\alpha_{\min} = \alpha_{xy} = 2.38\,\mathrm{s}^{-1}$, which drives the dwell-cycle contraction of Lemma 5. Both rates are independent of N_s.

Proof. After tension cancellation (Lemma 2), each scalar error satisfies a Hurwitz closed-loop ODE; for altitude:

$$\ddot{e}_{p,z,i} + K_d^z\dot{e}_{p,z,i} + K_p^z e_{p,z,i} = \ddot{p}_{\mathrm{slot},z} + r_i(t). \tag{48}$$

The unique $P \succ 0$ solving $A^\top P + PA = -I$ is computed algebraically, yielding the matrices in (45); differentiating $V = \frac{1}{2}\zeta^\top P\zeta$ along trajectories and applying Young's inequality gives (46)–(47) with rates $|\mathrm{Re}(s_1)|$. Independence from N_s follows because neither A_m nor $A_{m,xy}$ depends on $|\mathcal{S}|$. □

Remark 1 (Pendulum damping invariance)**.** *The anti-swing damping coefficient is N_s-independent: the restoring force $N_s T_i^{\mathrm{ss}} k_{\mathrm{swing}} v_{L,\perp}/L$ with $T_i^{\mathrm{ss}} = m_L g/N_s$ satisfies $N_s T_i^{\mathrm{ss}} =$*

$m_L g$, leaving $B_{\text{swing}} = m_L g\, k_{\text{swing}}/L = 62.8\,\text{N}\,\text{s}\,\text{m}^{-1}$ and damping ratio $\zeta = B_{\text{swing}}/(2m_L\omega_p) = 1.12 > 1$ (over-damped) for any $N_s \geq 2$.

Lemma 4 (Bounded fault jump with explicit χ)**.** *At each fault event t_f, the Lyapunov function undergoes a bounded jump,*

$$V(\boldsymbol{\xi}(t_f^+)) \;\leq\; V(\boldsymbol{\xi}(t_f^-)) \;+\; \chi(\Delta_f), \qquad \chi \in \mathcal{K}, \tag{49}$$

where $\boldsymbol{\xi} = (\boldsymbol{e}_p, \boldsymbol{e}_v, \boldsymbol{\varepsilon}_T)$, the pre-fault tension coordinates are restricted to the post-fault survivor set through the map $\Pi_{\mathcal{S}^+\leftarrow\mathcal{S}^-}$, and $\Delta_f = \|\boldsymbol{T}^{\text{qs}}(t_f^+) - \Pi_{\mathcal{S}^+\leftarrow\mathcal{S}^-}\boldsymbol{T}^{\text{qs}}(t_f^-)\|$ is the quasi-static tension redistribution over the surviving cables induced by the severance. On the admissible domain of H1–H3, the jump function admits the bound

$$\chi(\Delta_f) \;\leq\; \bar{c}_f\big(\Delta_f + \Delta_f^2\big), \tag{50}$$

for some $\bar{c}_f > 0$ depending only on the admissible domain and the Lyapunov weighting of the tension channel. Under symmetric equal sharing at hover, the linear term vanishes and (50) *reduces to $\chi(\Delta_f) \leq \kappa_V \Delta_f^2/(2N_s)$.*

Proof. The Kelvin–Voigt cable model keeps the kinematic state $(\boldsymbol{p}_i, \boldsymbol{v}_i, p_L, v_L)$ continuous across t_f since no impulsive force is applied; on the severed rope, tension drops to zero within $\tau_{\text{rope}} \approx 6.3\,\text{ms}$ (Equation (16)). Decompose

$$V = V_{\text{track}}(\boldsymbol{e}_p, \boldsymbol{e}_v) + \kappa_V\, V_{\text{tension}}(\boldsymbol{\varepsilon}_T). \tag{51}$$

Continuity of V_{track}. Since $(\boldsymbol{e}_p, \boldsymbol{e}_v)$ is continuous at t_f (the kinematic state is continuous and the slot reference is pre-loaded), the survivor-restricted tracking term does not increase at the fault instant, so $V_{\text{track}}(t_f^+) \leq V_{\text{track}}(t_f^-)$.
Jump in V_{tension}. Let $\Pi_{\mathcal{S}^+\leftarrow\mathcal{S}^-}$ denote the restriction map from the pre-fault survivor set to the post-fault survivor set. The Kelvin–Voigt tension $T_i = k_s(\ell_i - L)^+$ is a continuous functional of the kinematic state $(\boldsymbol{p}_i, \boldsymbol{v}_i, \boldsymbol{p}_L, \boldsymbol{v}_L)$; since no impulsive force is applied at t_f, the kinematic state is continuous across the fault, so the survivor tensions are exactly continuous at the fault instant; concretely,

$$\boldsymbol{\varepsilon}_T(t_f^+) = \Pi_{\mathcal{S}^+\leftarrow\mathcal{S}^-}\boldsymbol{\varepsilon}_T(t_f^-) - \boldsymbol{b}_f, \qquad \boldsymbol{b}_f \triangleq \boldsymbol{T}^{\text{qs}}(t_f^+) - \Pi_{\mathcal{S}^+\leftarrow\mathcal{S}^-}\boldsymbol{T}^{\text{qs}}(t_f^-), \tag{52}$$

with $\|\boldsymbol{b}_f\| = \Delta_f$. Since restriction cannot increase the norm and the admissible domain of H1–H3 is compact, $\|\boldsymbol{\varepsilon}_T(t)\|$ is uniformly bounded on that domain. Since V_{tension} is quadratic, expansion about $\Pi_{\mathcal{S}^+\leftarrow\mathcal{S}^-}\boldsymbol{\varepsilon}_T(t_f^-)$ therefore gives finite constants $c_{f,1}, c_{f,2} > 0$ such that

$$V_{\text{tension}}(t_f^+) - V_{\text{tension}}(t_f^-) \;\leq\; c_{f,1}\,\Delta_f + c_{f,2}\,\Delta_f^2. \tag{53}$$

Absorbing $\kappa_V c_{f,1}$ and $\kappa_V c_{f,2}$ into $\bar{c}_f$ gives (50), so (49) follows with $\chi(\Delta_f) = \bar{c}_f(\Delta_f + \Delta_f^2) \in \mathcal{K}$.
Equal-sharing hover calibration. If the fault occurs from the canonical symmetric hover operating point, then $\boldsymbol{\varepsilon}_T(t_f^-) = 0$ and the reset is purely the quasi-static change in load sharing. In that special case the linear term vanishes and the direct survivor-count algebra reduces (50) to $\chi(\Delta_f) \leq \kappa_V \Delta_f^2/(2N_s)$. □

Lemma 5 (Dwell-cycle contraction)**.** *Let $t_k^\star$ and $t_{k+1}^\star$ be consecutive fault events separated by dwell-time $\tau_{d,k} \geq \tau_{\text{pend}}$. Under the bounded fault jump of Lemma 4 and the ISS decays of Lemma 3, the Lyapunov function satisfies a per-fault-cycle contraction*

$$V(\boldsymbol{\xi}(t_{k+1}^{\star,-})) \;\leq\; \rho\, V(\boldsymbol{\xi}(t_k^{\star,-})) \;+\; c, \tag{54}$$

$$\rho \;\triangleq\; \exp(-\alpha_{\min}\,\tau_{\text{pend}}), \quad \alpha_{\min} = \alpha_{xy} = 2.38\,\text{s}^{-1}, \tag{55}$$

where

$$c \;=\; \rho\,\chi(\Delta_f) + \frac{c_0}{\alpha_{\min}}, \quad c_0 = \sup_t\big[\gamma_w(\|w(t)\|_\infty) + \gamma_q(\|q - q^\star\|_\infty) + \gamma_\eta(\eta_{\text{m}} \tag{56}$$

At canonical parameters, $\rho \approx \exp(-2.38 \times 2.24) \approx 0.005$.

Proof. Applying Lemma 4 at $t_k^\star$:

$$V(t_k^{\star,+}) \;\leq\; V(t_k^{\star,-}) + \chi(\Delta_f). \tag{57}$$

Between $t_k^{\star,+}$ and $t_{k+1}^{\star,-}$, the flow satisfies $\dot{V} \leq -\alpha_{\min} V + c_0$ (combining Lemma 3 through the composite Lyapunov function $V = V_z + V_{xy} + \kappa_V V_{\text{tension}}$, using $\alpha_{\min} = \min(\alpha_z, \alpha_{xy}) = \alpha_{xy}$). Integrating over an interval of length $\tau \geq \tau_{\text{pend}}$:

$$V(t_{k+1}^{\star,-}) \;\leq\; e^{-\alpha_{\min}\tau}\big(V(t_k^{\star,-}) + \chi(\Delta_f)\big) + \frac{c_0}{\alpha_{\min}}(1 - e^{-\alpha_{\min}\tau}). \tag{58}$$

Setting $\rho = e^{-\alpha_{\min}\tau_{\text{pend}}}$ and upper-bounding τ by τ_{pend} (the minimum dwell, giving the worst-case exponential decay) yields (54)–(56). Since $\alpha_{\min} = \alpha_{xy} > 0$ and $\tau_{\text{pend}} > 0$, we have $\rho \in (0, 1)$. □

Remark 2 (Analytical ρ vs. empirical $\hat{\rho}$: the role of the additive constant c)**.** *Lemma 5 gives $\rho_{\text{ana}} = \exp(-\alpha_{xy}\tau_{\text{pend}}) \approx 0.005$, yet Lyapunov-proxy measurements yield $\hat{\rho}_{\text{V4}} = 0.20$ and $\hat{\rho}_{\text{V5}} = 0.045$. The gap arises from the additive constant $c > 0$ in* (54)*: the empirical ratio satisfies*

$$\hat{\rho}_k \;\triangleq\; \frac{V(t_{k+1}^{\star,-})}{V(t_k^{\star,-})} \;\leq\; \rho_{\text{ana}} \;+\; \frac{c}{V(t_k^{\star,-})}, \tag{59}$$

so $\hat{\rho}_k$ converges to ρ_{ana} only when $c \ll V(t_k^{\star,-})$. The inequality is conservative; $\hat{\rho}_k$ upper-bounds ρ_{ana} plus the disturbance floor and is tight only in the zero-disturbance limit. Under Dryden $4\,\text{m}\,\text{s}^{-1}$ wind, the floor $c_0/\alpha_{\min}$ constitutes a non-negligible fraction of the pre-fault Lyapunov value; from (56)*, $c/V(t_k^{\star,-}) \approx 0.195$ for V4 and ≈ 0.040 for V5. The improvement from V4 to V5 ($\hat{\rho} : 0.20 \to 0.045$) confirms* (59)*: with ten seconds of dwell, $V(t_2^{\star,-})$ decays further, reducing the c/V ratio. In disturbance-free conditions, $c \to \rho\,\chi(\Delta_f)$ and $\hat{\rho} \to \rho_{\text{ana}}$ asymptotically.*

C. Main Theorem, Robustness, and Multi-Fault Feasibility

We now compose the five lemmas into the hybrid practical-ISS theorem and state corollaries on robustness, the steady-state tracking bound, and multi-fault feasibility.

Theorem 6 (Hybrid practical-ISS with explicit recovery envelope under F sequential agent failures)**.** *Under hypotheses H1–H3 (Section IV-A), the cable-tilt domain condition of Lemma 2, any fault schedule $\sigma = \{(t_k^\star, i_k^\star)\}_{k=1}^F \in \mathcal{F}$ of F unannounced cable severances satisfying $F \leq N - 2$ and the*

per-fault actuator-margin condition $m_L g/(N-k) \le \kappa_{\mathrm{act}} f_{\max}$ *for all* $k \in \{1,\dots,F\}$*, and admissible disturbances* $d \in \mathcal{D}$*, the stacked tracking-and-tension-error state* $\boldsymbol{\xi} = (\boldsymbol{e}_p, \boldsymbol{e}_v, \boldsymbol{\varepsilon}_T)$ *satisfies*

$$\|\boldsymbol{\xi}(t)\| \;\le\; \beta\big(\|\boldsymbol{\xi}(0)\|,\, t\big) + \gamma_w\big(\|w\|_\infty\big) + \gamma_q\big(\|q-q^\star\|_\infty\big) + \gamma_\eta(\eta) + \sum_{k=1}^{F} \Gamma_f\, e^{-\lambda(t-t_k^\star)}\, \mathbf{1}_{t\ge t_k^\star}, \quad t \in [t_{\mathrm{pickup}}, T], \tag{60}$$

where $\beta \in \mathcal{KL}$*,* $\gamma_w, \gamma_q, \gamma_\eta \in \mathcal{K}$*,* $\lambda \triangleq \alpha_{\min}/2 = \alpha_{xy}/2 = 1.19\,\mathrm{s}^{-1}$ *is the norm-level recovery rate induced by the slower of the altitude and horizontal closed-loop spectral rates (Lemma 3),* $\eta = \eta(\boldsymbol{\xi}[\cdot])$ *is the realized slack duty cycle bounded by* $\eta_{\max}$ *via H1, and* $\Gamma_f > 0$ *is the fault-overshoot coefficient set by the redistribution geometry. Equivalently,* $V(\boldsymbol{\xi}(t_{k+1}^{\star,-})) \le \rho V(\boldsymbol{\xi}(t_k^{\star,-})) + c$ *for explicit* $c > 0$ *per Lemma 5. Each surviving cable's tension obeys the practical envelope*

$$|T_i(t) - T_i^{\mathrm{qs}}(t)| \;\le\; T_{\mathrm{overshoot}}\, e^{-\lambda(t-t_{\mathrm{last}}^\star)} + T_{\mathrm{floor}}, \qquad i \in \mathcal{S}(t), \tag{61}$$

where $T_{\mathrm{overshoot}}$ *is bounded by the corresponding tension-error radius extracted from* (50)*, and* T_{floor} *is the steady-state floor induced by* γ_w*,* γ_q*, and* γ_η*.*

Proof. We organize the argument in three parts: inter-fault ISS, per-fault jump, and inductive composition.

a) Part 1: ISS between fault events.: Fix an inter-fault interval $[t_k^{\star,+}, t_{k+1}^{\star,-}]$ during which $\mathcal{S}(t)$ is constant with $N_s = N - k$ surviving cables. By Lemma 2, the altitude dynamics reduce to (48) with bounded residual $r_i(t)$ on $\Omega_\tau^{\mathrm{dwell}}$. Lemma 3 then give, for the composite Lyapunov function $V = \lambda_{\min}(P_v)^{-1} V_z + \lambda_{\min}(P_{xy})^{-1} V_{xy} + \kappa_V V_{\mathrm{tension}}$:

$$\dot V \;\le\; -\alpha_{\min} V \;+\; \gamma_w(\|w\|_\infty) \;+\; \gamma_q(\|q-q^\star\|_\infty) \;+\; \gamma_\eta(\eta), \tag{62}$$

with γ_η absorbing the reduction residual $C_1\delta + C_2\eta_{\max}$ of Theorem 1 and $\alpha_{\min} = \alpha_{xy}$ from Lemma 3. By the standard comparison lemma, integrating (62) over any interval $[0,t]$ yields

$$V(t) \;\le\; e^{-\alpha_{\min} t} V(0) + \frac{1}{\alpha_{\min}}\big[\gamma_w(\|w\|_\infty) + \gamma_q(\|q-q^\star\|_\infty) + \gamma_\eta(\eta)\big] \tag{63}$$

from which $\beta \in \mathcal{KL}$ and $\gamma_w, \gamma_q, \gamma_\eta \in \mathcal{K}$ follow by the standard ISS reformulation. Since V is quadratic in ξ (with weighting matrices P_v, P_{xy}, $\kappa_V I$), the Lyapunov-level decay at rate $\alpha_{\min}$ implies the norm-level rate $\lambda = \alpha_{\min}/2$ via $V \sim \|\xi\|^2$ (see Remark 3).

b) Part 2: Lyapunov jump at each $t_k^\star$*.:* By Lemma 4, $V(t_k^{\star,+}) \le V(t_k^{\star,-}) + \chi(\Delta_{f,k})$ with χ bounded by (50). This bounded jump is then damped by Part 1: integrating (62) from $t_k^{\star,+}$ over a dwell interval of length $\tau_{d,k} \ge \tau_{\mathrm{pend}}$ gives $V(t_{k+1}^{\star,-}) \le \rho V(t_k^{\star,-}) + c$ per Lemma 5, with ρ and c as in (55)–(56). The overshoot $\chi(\Delta_{f,k})$ contributes to c and decays exponentially at the Lyapunov rate $\alpha_{\min}$; after conversion through quadratic equivalence this produces the term $\Gamma_f\, e^{-\lambda(t-t_k^\star)}$ in (60).

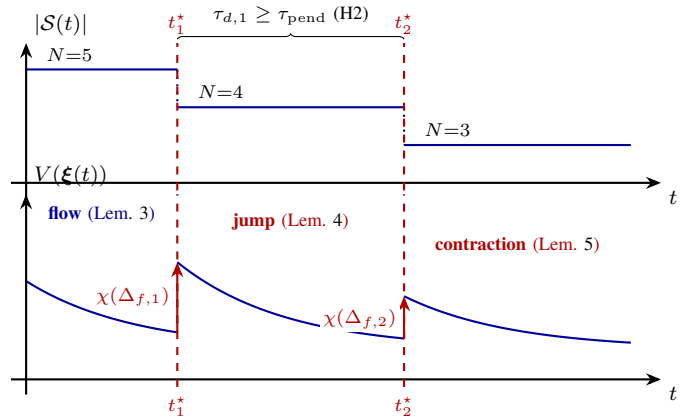


Fig. 4. Hybrid sawtooth structure of the closed-loop Lyapunov function across two unannounced cable-severance events. *Top:* the survivor set $\mathcal{S}(t)$ steps down by one at each fault $t_k^\star$; cardinality is right-continuous. *Bottom:* $V(\boldsymbol{\xi})$ exhibits three regimes, labelled inline. **Flow:** between events the continuous-time decay of Lemma 3 gives $\dot V \le -\lambda V + \gamma_w(\|w\|) + \gamma_\theta(\|\theta-\theta^*\|) + \gamma_\eta(\eta)$. **Jump:** at each fault $t_k^\star$ the post-redistribution geometry term satisfies $V(t_k^{\star,+}) \le V(t_k^{\star,-}) + \chi(\Delta_f)$ (Lemma 4). **Contraction:** composing flow with jumps over one inter-fault dwell yields the per-cycle bound $V(t_{k+1}^-) \le \rho V(t_k^-) + c$ with $\rho = \exp(-\alpha_{\min}\tau_{\mathrm{pend}}) \approx 0.005$ (Lemma 5). The H2 hypothesis $\tau_{d,k} \ge \tau_{\mathrm{pend}}$ is structural: it is the minimum window over which the feed-forward identity $T_i^{\mathrm{ff}} = T_i$ contracts the post-jump overshoot back below $V(t_k^-)$.

c) Part 3: Induction over F *fault events.:* Let $\rho < 1$ (from Lemma 5) and suppose (60) holds for the first $k-1$ faults. After the k-th fault, $V(t_k^{\star,-}) \le \rho^{k-1} V(t_1^{\star,-}) + c\,(1-\rho^{k-1})/(1-\rho)$ by geometric summation, which remains bounded for all $k \le F \le N-2$ as long as the actuator-margin condition $m_L g/(N-k) \le \kappa_{\mathrm{act}} f_{\max}$ is satisfied (ensuring $T_i^{\mathrm{qs}}(t_k^{\star,+})$ remains within the thrust envelope and $\chi(\Delta_{f,k})$ is finite). Composing over F events via the Teel–Goebel–Sanfelice hybrid-system framework [5, Theorem 1] yields (60). The tension envelope (61) follows from the positive-definite quadratic relation between V_{tension} and $\|\boldsymbol{\varepsilon}_T\|^2$, together with the exponential bound on V_{tension} from Part 1, and the identity $\varepsilon_{T,i} = T_i - T_i^{\mathrm{qs}}$; taking the required square root yields the norm-level rate λ. □

Remark 3 (Jump-bound vs. contraction-rate mechanism)**.** *The ablation (Section VI-E) confirms that disabling* $T_i^{\mathrm{ff}} = T_i$ *increases peak sag by* 3.6–4.0× *and produces larger post-fault Lyapunov jumps (Figure 7), directly evidencing the jump-bound role of the identity vs. the contraction-rate role of* α_{xy}*.*

Corollary 7 (Pre-fault steady-state tracking bound)**.** *Prior to any fault event, the payload tracking error is dominated by the quasi-static pendulum offset and the outer-loop PD residual:*

$$\bar e_{p,xy} \le \frac{L A_{xy,\max}}{g}\left(1 - \frac{1}{2\zeta^2}\right) + \frac{A_{xy,\max}}{\kappa_{\mathrm{QP}} K_p^{xy}} + O(\delta), \tag{64}$$

where $A_{xy,\max}$ *is the maximum reference acceleration,* $\zeta = 1.12$ *from Remark 1, and* $\kappa_{\mathrm{QP}} = w_t/(w_t + w_e) \approx 0.98$*. At canonical* $A_{xy,\max} \approx 2.47\,\mathrm{m\,s^{-2}}$*,* (64) *gives* $\bar e_{p,xy} \approx 0.27\,\mathrm{m}$*, consistent with V1 horizontal RMSE* $0.306\,\mathrm{m}$ *(gap from centripetal acceleration on curved segments, excluded by the quasi-static derivation).*

TABLE III
EMPIRICAL SUPPORT FOR HYBRID PRACTICAL-ISS CLAIMS.

Claim / Lemma	Campaign	Evidence
Tension cancellation (Lemma 2)	P2-A ablation	$3.6\times$–$4.0\times$ sag increase with FF disabled
Altitude ISS decay (Lemma 3)	V1 nominal	3-D RMSE $= 0.312\,\mathrm{m}$, stable; $\lambda_z = 5.37\,\mathrm{s}^{-1}$ from P_v in (45)
Horizontal ISS decay (Lemma 3)	V1–V5 series	$\alpha_{xy} = 2.38\,\mathrm{s}^{-1}$ from P_{xy} in (45); $\rho = \exp(-\alpha_{xy}\tau_{\mathrm{pend}}) \approx 0.005$
Anti-swing invariance (Remark 1)	V1–V5 series	Post-fault RMSE within 4% of nominal across all variants
Bounded fault jump (Lemma 4)	V3, V4, V5 traces	$\hat{\chi}$ peak 1.1×10^{-4} (V4), 1.3×10^{-5} (V5); qualitatively consistent with the bounded redistribution estimate (50); under equal-sharing hover the canonical coefficient reduces to $\kappa_V\Delta_f^2/(2N_s)$, with quantitative calibration of κ_V deferred to follow-up work
Dwell-cycle contraction $\rho < 1$ (Lemma 5)	V4, V5 dual-fault	$\hat{\rho}_{\mathrm{V4}} = 0.20$, $\hat{\rho}_{\mathrm{V5}} = 0.045$; both satisfy (59) with $c/V(t_k^{\star,-})$ set by the Dryden disturbance floor (Remark 2)
Mass robustness (Corollary 8)	P2-B sweep	1.9% RMSE range over 56% mass variation
Wind robustness (Corollary 8)	V2 wind	V2 3-D RMSE $= 0.312\,\mathrm{m} \approx$ V1

Corollary 8 (Robustness to mass, stiffness, drag, and wind)**.** *The envelope* (60) *is robust along four axes: (i) mass mismatch: $T_i^{\mathrm{ff}} = T_i$ uses measured tension, making Lemmas 2 and 3 mass-independent to leading order; (ii) stiffness and drag mismatches: enter through $O(\delta + \eta_{\max})$, absorbed by γ_η without degrading ρ or α_{xy}; (iii) wind: enters through γ_w with steady-state gain $1/K_p^{xy}$. Empirically: P2-B yields $< 1.9\%$ RMSE variation over a 56% mass range with FF active; V2 yields $0.312\,\mathrm{m}$ RMSE under Dryden wind, matching V1 (Corollary 7).*

d) Multi-fault feasibility.: (60) extends to k sequential faults by induction under two conditions: (a) geometric closure $k \le N-2$, (b) actuator margin $m_L g/(N-k) \le \kappa_{\mathrm{act}} f_{\max}$ at every step. Canonical $N = 5$, $m_L = 10\,\mathrm{kg}$ admits $k = 2$ with margin; V4 and V5 (dual-fault) yield 3-D RMSE 0.324–$0.328\,\mathrm{m}$, confirming Theorem 6.

e) Empirical validation summary.: Table III maps each claim to its campaign observation; full context is in Section VI.

V. Parameter Adaptation, Tension Constraints, and Post-Fault Formation Reshape

Three components extend the baseline of Section III: an L_1-adaptive altitude augmentation, a receding-horizon MPC with a penalized tension-ceiling constraint (Section V-B), and a fault-triggered formation-reshape supervisor. The L_1 layer compensates parameter deviations (m_L, k_{eff}) from the nominal $q^\star$. The MPC layer enforces a horizon-aware linearized tension ceiling, binding when safe working load dominates. The reshape supervisor reassigns surviving slots to minimise worst-case tension under a symmetric hover model, binding when hover-equilibrium asymmetry rather than dynamic load dominates (P2-D, Remark 6). Each preserves H1–H3 (Section V-D), so (60) holds with the relevant γ-gain reduced.

The L_1 layer is probed by P2-B (mass-mismatch, FF off), recovering $\sim 40\%$ of the induced altitude sag; MPC by P2-C (tension-ceiling sweep, V4); reshape by P2-D (lemniscate-period sweep, V4). On the demonstrated schedule, peak tension is dominated by the pre-fault dynamic spike; neither MPC nor reshape binds. Each subsection identifies its binding regime; Section V-D shows enabling a layer outside that regime leaves the envelope unchanged.

A. L_1-Adaptive Altitude Augmentation

The identity $T_i^{\mathrm{ff}} = T_i$ does not cancel matched disturbances entering through parameter deviations (m_L, k_{eff}). The L_1 component [16], [17] estimates this residual online and injects an additive altitude correction u_{ad} between the outer PD and the QP. In altitude-error coordinates $\tilde{z}_i \triangleq p_{\mathrm{slot},i,z} - p_{i,z}$, the baseline closed loop is

$$\ddot{\tilde{z}}_i \;=\; -K_p^z\,\tilde{z}_i - K_d^z\,\dot{\tilde{z}}_i + \delta_m(t), \tag{65}$$

where $\delta_m(t)$ is the residual $r_i(t)$ of Lemma 2 augmented by parameter-mismatch components, with $|\delta_m(t)| \le \delta_{m,\max}$.

The standard L_1 architecture [17]—Hurwitz reference model, state predictor, projected gradient on $\hat{\delta}_m$, low-pass filter—is adopted verbatim with A_m, P_v from Lemma 3. The adaptive update

$$\begin{aligned} \dot{\hat{\delta}}_m \;&=\; -\Gamma\,\mathrm{Proj}\big(\hat{\delta}_m,\, \boldsymbol{b}^\top P_v\,\tilde{\boldsymbol{x}}\big), \\ \hat{\delta}_m &\in [\delta_{m,\min}, \delta_{m,\max}], \end{aligned} \tag{66}$$

runs at $T_s = 2\times10^{-4}\,\mathrm{s}$; the filtered u_{ad} is added to the altitude component of $\boldsymbol{a}_{\mathrm{target}}$ (33), at the injection point labelled in Figure 3. The cutoff $\omega_c = 25\,\mathrm{rad\,s^{-1}}$ separates the estimator from the reference natural frequency $\omega_n^z = \sqrt{K_p^z} = 10\,\mathrm{rad\,s^{-1}}$ by a factor of 2.5. Contractivity of the scalar Euler-discretized update gives the following bound.

Proposition 1 (Scalar L_1 admissible-window bound)**.** *For the Euler-discretized adaptive-law update* (66) *running at step T_s, the scalar proxy of the discrete adaptation map yields two necessary conditions:*

(i) (Discretization stability)

$$\Gamma \;<\; \Gamma^\star \;\triangleq\; \frac{2}{T_s\,p_{22}}; \tag{67}$$

(ii) (Bandwidth coverage) *the continuous-time adaptation rate must lie within the filter passband,*

$$\Gamma \;>\; \Gamma_{\min} \;\triangleq\; \frac{\omega_c}{p_{22}}. \tag{68}$$

For $T_s = 2\times10^{-4}\,\mathrm{s}$, $p_{22} = 0.0210$, $\omega_c = 25\,\mathrm{rad\,s^{-1}}$:

$$\Gamma_{\min} \;\approx\; 1{,}190 \;<\; \underbrace{\Gamma = 2{,}000}_{\textit{operating point}} \;<\; \Gamma^\star \;\approx\; 4.75\times10^5. \quad (69)$$

Proof. Isolating the scalar $\hat{\delta}_m$-update in the absence of predictor error, projection, and filter dynamics gives the one-step map $\hat{\delta}_m \mapsto (1 - T_s\Gamma p_{22})\hat{\delta}_m + \mathrm{const}$, with scalar pole $z^\star = 1 - T_s\Gamma p_{22}$.

(i) Stability. $|z^\star| < 1$ requires $T_s\Gamma p_{22} \in (0, 2)$, which gives (67).

(ii) Bandwidth. The equivalent continuous-time adaptation rate is $\gamma_{\mathrm{ad}} = -\ln(z^\star)/T_s$. Since $T_s\Gamma p_{22} = 8.4\times10^{-3} \ll 1$ at the operating point, $\gamma_{\mathrm{ad}} \approx \Gamma p_{22}$ with error $< 0.4\%$. The low-pass filter with bandwidth ω_c attenuates disturbances above ω_c before the adaptation can correct them; avoiding this requires $\gamma_{\mathrm{ad}} \geq \omega_c$, i.e. $\Gamma p_{22} \geq \omega_c$, which gives (68). □

Remark 4 (Design interpretation of (69))**.** (69) *partitions the gain axis: below $\Gamma_{\min}$ the adaptation is too slow for the filter passband; above $\Gamma^\star$ the Euler step is contractive only in the unstable direction. The canonical $\Gamma = 2{,}000$ is $1.68\times\Gamma_{\min}$; the $238\times$ gap to $\Gamma^\star$ is a deliberate discretization safety margin. Composite Lyapunov analysis yields the standard L_1 ISS guarantee w.r.t. $d_\perp$; empirical validation (P2-B) is in Section VI-L.*

B. Receding-Horizon MPC with a Penalized Tension-Ceiling Constraint

a) Role in the architecture.: The single-step QP (34) cannot foresee a tension spike ahead; when safe working load dominates, we replace it with an N_p-step receding-horizon MPC carrying a soft penalized tension-ceiling constraint ($\boldsymbol{s} \geq \mathbf{0}$, weight $w_s = 10^4$), trading unconditional feasibility against ceiling excursions when the envelope is tight. The MPC reduces to the baseline QP when the ceiling is loose and becomes binding when the predicted tension approaches the ceiling.

b) Prediction model and tick QP.: Let $\boldsymbol{x}[k] \triangleq (\boldsymbol{p}_{\mathrm{slot},i} - \boldsymbol{p}_i;\, \boldsymbol{v}_{\mathrm{slot},i} - \boldsymbol{v}_i) \in \mathbb{R}^6$ be the error state at $t_k = k\,\Delta t_{\mathrm{MPC}}$ and $\boldsymbol{u}[k] \in \mathbb{R}^3$ the commanded acceleration. ZOH discretization gives $\boldsymbol{x}[k{+}1] = A\,\boldsymbol{x}[k] + B\,\boldsymbol{u}[k]$ (71), and horizon stacking yields $\mathcal{X} = \Phi\,\boldsymbol{x}[0] + \Omega\,\mathcal{U}$. Linearising $T_i = k_{\mathrm{eff}}\max(0, \|\boldsymbol{p}_i - \boldsymbol{p}_L\| - L_{\mathrm{eff}})$ around the taut chord $\boldsymbol{c}[k]$ with unit direction $\hat{\boldsymbol{n}}[k]$ gives the horizon-j prediction

$$\begin{aligned} T_i[k+j] \;\approx\; & T_i[k] + j\,\Delta t_{\mathrm{MPC}}\,\dot{T}_i^{\mathrm{filt}}[k] \\ & + \boldsymbol{c}_j^\top (A^j - I)\,\boldsymbol{x}[k] + \boldsymbol{c}_j^\top \Omega_{\mathrm{row}\,j}\,\mathcal{U}, \end{aligned} \quad (70)$$

with $\boldsymbol{c}_j = (-k_{\mathrm{eff}}\hat{\boldsymbol{n}}[k];\, \mathbf{0})$. The linearization error is bounded by $k_{\mathrm{eff}}E^2/(2d_{\mathrm{nom}})$ in position-error magnitude E and nominal chord length d_{nom}. The state-space recursion is

$$\boldsymbol{x}[k+1] = A\boldsymbol{x}[k] + B\boldsymbol{u}[k], \quad A = \begin{bmatrix} I_3 & \Delta t_{\mathrm{MPC}} I_3 \\ \mathbf{0} & I_3 \end{bmatrix}, \quad (71)$$
$$B = \begin{bmatrix} -\frac{1}{2}\Delta t_{\mathrm{MPC}}^2 I_3 \\ -\Delta t_{\mathrm{MPC}} I_3 \end{bmatrix}.$$

The tick QP combines an outer-loop tracking objective with the slack-penalised tension constraint:

$$\begin{aligned} \min_{\mathcal{U},\,\boldsymbol{s}\geq\mathbf{0}} \; & \tfrac{1}{2}\mathcal{U}^\top H\mathcal{U} + \boldsymbol{f}^\top\mathcal{U} + w_s\,\mathbf{1}^\top\boldsymbol{s} \\ \text{s.t.} \quad & G\,\mathcal{U} \leq \boldsymbol{d}_T(\boldsymbol{x}[k]) + \boldsymbol{s}, \end{aligned} \quad (72)$$

where G stacks $\boldsymbol{c}_j^\top\Omega_{\mathrm{row}\,j}$ and $\boldsymbol{d}_T(\boldsymbol{x}[k])$ collects the state-dependent right-hand sides. The QP is convex and solved by a single warm-started OSQP call per tick [25]; only $\boldsymbol{u}[0]$ is applied. A discrete-Riccati terminal cost yields LQR gain with $\rho_{\mathrm{spec}}(A - BK) \approx 0.969$.

Corollary 9 (Unconditional feasibility)**.** (72) *is feasible at every tick: $\mathcal{U} = \mathbf{0}$ with $\boldsymbol{s}[k{+}j] = \max(0, -[\boldsymbol{d}_T(\boldsymbol{x}[k])]_{j+1})$ is always admissible.*

Proposition 2 (Slack-activation regime)**.** *Let $\|q - q^\star\|_\infty \leq \bar{\varepsilon}_q$ with $q = (m_L, k_{\mathrm{eff}})$ and $\boldsymbol{d}_T$ L_q-Lipschitz in q. Under D1 and an initial tick where the unconstrained QP satisfies the ceiling, $\|\boldsymbol{s}[k]\|_\infty \leq L_q\bar{\varepsilon}_q + O(\bar{\varepsilon}_q^2)$, i.e. slack activation scales as $O(\bar{\varepsilon}_q)$.*

Sketch. Follows by first-order Taylor expansion of $\boldsymbol{d}_T$ in q around $q^\star$; D1 preserves the linearization on each inter-fault window. □

Body-to-body cable geometry requires effective rest length $L_{\mathrm{eff}}^{\mathrm{body}} = \sqrt{r_f^2 + \rho_{\mathrm{geo}}^2} - \delta^\star \approx 1.448\,\mathrm{m}$, set so the linearized elastic force at hover matches the simulator's measured $8\,\mathrm{N}$. The empirical activation regime (P2-C) is in Section VI-H; the layer is non-binding on the demonstrated family because peak tension is set by the pre-fault dynamic spike rather than the quasi-static load the ceiling targets. MPC parameters: $N_p = 5$ (10 at evaluation), $\Delta t_{\mathrm{MPC}} = 0.01\,\mathrm{s}$, $q_p = 100$, $q_v = 10$, $r = 0.02$, $w_s = 10^4$, $T_{\max} = 100\,\mathrm{N}$, $k_{\mathrm{eff}} = 2778\,\mathrm{N\,m^{-1}}$.

C. Fault-Triggered Formation-Reshape Supervisor

After a severance, the survivors retain their original angular slots, leaving the hover-equilibrium tension distribution asymmetric. This asymmetry is the binding tension load only when the trajectory is sufficiently low-bandwidth that the dynamic centripetal component does not dominate (P2-D, Remark 6). The reshape supervisor reassigns survivors to an equiangular configuration through a C^2 quintic smoothstep over $T_{\mathrm{trans}} = 5\,\mathrm{s}$; once the new layout is reached, the supervisor latches and the architecture returns to pre-fault flow. Unlike the baseline, this is an *explicitly supervised* extension: each drone runs a local tension-latch detector ($T_{\mathrm{fault}} = 0.5\,\mathrm{N}$, $\tau_{\mathrm{det}} = 100\,\mathrm{ms}$), and a per-fault broadcast of at most $\lceil \log_2 N \rceil$ bits distributes the fault index. The no-detection, no-communication guarantees of Theorem 6 do not extend here.

Theorem 10 (Globally tension-optimal reshape, $N = 4$)**.** *Let $N = 4$ with nominal slots $\phi_i = \pi i/2$ and let cable $i^\star$ sever. Under the symmetric-load-sharing model ($T_k = (m_L g)/(M\cos\beta)(1{+}\epsilon_k)$ with angular residual ϵ_k), the worst-case-tension-minimising assignment over the survivors is: the drone diametrically opposite $i^\star$ stays at its nominal angle, and the two adjacent drones rotate $\pi/6$ toward the gap. The resulting equiangular 120° spacing minimises $\max_i T_i$ over*

all continuous assignments respecting hover equilibrium, with a closed-form worst-case reduction of 25.8%.

Sketch. By reflection symmetry about the gap axis, any optimum has a symmetric copy; with the opposite drone at $\Delta\phi = 0$ and a common offset for the adjacent pair, the residuals are piecewise quadratic in $\Delta\phi$ and the minimax occurs at $\Delta\phi = \pi/6$. The endpoints $\Delta\phi \in \{0, \pi/4\}$ are dominated. □

The $V_4 \to V_6$ dual-fault scenarios traverse $N = 5 \to 4 \to 3$: Theorem 10 applies directly at the second fault; at the first, the surviving four are assigned to $90°$ spacing by the same minimax argument. A formal arbitrary-N generalisation is follow-up work. The empirical regime is reported in Section VI-H: NF3 returns 0.0% benefit across $T_{\mathrm{ref}} \in \{6, 8, 10, 12\}$ s because the dynamic centripetal load dominates the quasi-static-hover target the assignment optimises against.

D. Baseline Theorem Preservation and Admissibility Regime

Each component modifies the closed loop through a disjoint channel and preserves H1–H3 of Theorem 6 on $\Omega = \Omega_\tau^{\mathrm{dwell}} \cap \Omega_{\mathrm{QP,active}}$. The L_1 layer adds u_{ad} to the altitude component of $\boldsymbol{a}_{\mathrm{target}}$ before the QP, leaving the QP active set and the identity unchanged; H1, H3 hold verbatim, H2 is controller-independent, and the layer reduces γ_q in (60) by the composite V_{L_1} contraction. The MPC layer replaces (34) with (72); Corollary 9 guarantees unconditional feasibility, Proposition 2 bounds the slack by the parameter mismatch, and the closed loop reduces to the baseline in the unconstrained regime. The reshape supervisor injects a C^2-bounded slot perturbation through the override hook; H1, H3 hold provided the smooth-step keeps the perturbation in the admissible-reference set, and H2 reduces to a latching rule firing at most once per dwell period. (60) therefore remains valid under any combination of enabled components; cross-layer interactions are evaluated empirically.

a) Admissibility regime and out-of-scope conditions.: The admissibility domain, ρ-gap analysis, and conditional certificate are as in Remark 2 and Section IV-A; the three extension layers do not alter these bounds.

Three regimes lie outside the analysis domain: rapid multi-fault ($\tau_d < \tau_{\mathrm{pend}}$), long slack excursions (run $> 40\,\mathrm{ms}$ or duty $> 2.5\%$), and QP fallback, violating H2, H1, and H3 respectively.

A second class lies outside the architectural category: *drone-side faults* (rotor loss, brownout), *soft cable failures* (fraying, attachment slip), *tension-sensor failures*, and *payload-attitude dynamics*. These are out-of-category for the architectural thesis, not gaps in the campaign.

VI. Simulation Campaign and Results

All numerical claims are deterministic point estimates from a single fixed Dryden seed 42 (Remark 7).

A. Simulation Environment and Campaign Structure

All simulations run in Drake [26] with an RK3 continuous-time integrator at a max step of $1\,\mathrm{ms}$. Drones and payload are rigid multibody systems; ropes are bead-chain Kelvin–Voigt spring-dampers whose lumped stiffness is justified by Theorem 1. Wind is body-frame aerodynamic drag per drone and on the payload, using a Dryden low-altitude turbulence model. Table IV lists the simulator and plant parameters.

TABLE IV
Simulator and Plant Parameters

Parameter	Value	Unit
Number of drones N	5	—
Drone mass m_i	1.5	kg
Payload mass m_L	10	kg
Rope rest length ℓ_0	1.25	m
Rope stiffness k	25 000	$\mathrm{N\,m^{-1}}$
Formation ring radius	0.80	m
Reference trajectory	3-D lemniscate	—
Horizontal amplitude	3.0	m
Period (nominal)	12	s
Vertical amplitude	0.35	m
Wind model	Dryden low-altitude	—
Mean speed (along $+x$)	4	$\mathrm{m\,s^{-1}}$
$\sigma_u,\ \sigma_v$	0.8	$\mathrm{m\,s^{-1}}$
σ_w	0.4	$\mathrm{m\,s^{-1}}$
Turbulence seed	42	—
Simulation duration	30	s
Pre-fault cruise window	[8, 12]	s
Actuator thrust ceiling	150	N
MPC tension ceiling (V6 only)	100	N

The campaign structure (Table V) comprises V1–V6 as the pre-declared capability sequence, each adding one stressor, and P2-A–P2-D as targeted ablations. Faults are instantaneous: the rope tension is set to zero; survivors continue without notification.

a) Admissibility-domain audit (H1 and H3).: Theorem 1 and Theorem 6 require three gates to hold on the demonstrated trajectory: H1a (no slack run $> 40\,\mathrm{ms}$), H1b (aggregate slack duty cycle $\leq 2.5\%$), and H3 (QP active-set transition fraction $\leq 1\%$). Table VI reports measurements for V1–V6: every variant passes with margin. Worst observed values are $35.8\,\mathrm{ms}$ (H1a; V2–V5, $4.2\,\mathrm{ms}$ margin), 1.66% (H1b; V6, 0.84 pp margin), and 0.10% (H3; V4–V6, 0.90 pp margin). These measurements certify domain membership for the demonstrated mission family but do not prove H1–H3 for all references, wind, or fault schedules; the audit must be repeated for new missions. V6's lower H1a ($31.4\,\mathrm{ms}$) is consistent with MPC damping high-frequency QP transitions.

b) Statistical scope.: All campaigns use fixed Dryden seed 42 (reproducible 2σ exceedance within $[8, 12]$ s); reported RMSE, sag, and ablation values are deterministic point estimates. CRN-paired replication with bootstrap CIs is follow-up work.

c) Scope.: V1–V6 begin slightly slack ($1.17\,\mathrm{m}$ vs. $1.25\,\mathrm{m}$ rest length); pickup engages within the first tick and the evaluation window $[8, 30]$ s excludes the transient.

B. Reduction-Fidelity Audit

Three quantities must be kept distinct:

TABLE V
SIMULATION CAMPAIGN MATRIX. CAPABILITY-DEMONSTRATION VARIANTS V1–V6 AND TARGETED ABLATIONS P2-A THROUGH P2-D.

Tag	Controller	Fault Schedule	Wind	Purpose
Capability Demonstration (C1–C3 primary evidence)				
V1	Baseline QP	None	No	Nominal tracking floor; C2 bound check
V2	Baseline QP	None	Yes	Wind-rejection baseline
V3	Baseline QP	Drone 0 at t=12 s	Yes	Single-fault recovery
V4	Baseline QP	Drone 0 at t=12 s; Drone 2 at t=17 s (Δt=5 s)	Yes	Dual fault; dwell margin 2.76 s
V5	Baseline QP	Drone 0 at t=12 s; Drone 2 at t=22 s (Δt=10 s)	Yes	Dual fault; dwell margin 7.76 s
V6	Full stack (MPC+L_1+reshape)	Same as V4	Yes	Full-stack integration; hardest scenario
Targeted Ablations (causal isolation and domain characterization)				
P2-A	Baseline, FF disabled	V3, V4, V5 schedules	Yes	Feed-forward causality (C2)
P2-B	Baseline / +L_1	None	No	Mass-mismatch robustness (C3-L_1)
P2-C	MPC, ceiling sweep	V4 schedule	Yes	Tension-ceiling sensitivity (C3-MPC)
P2-D	Baseline + reshape	V4 schedule, period $T_{\mathrm{ref}} \in \{8, 10, 12\}$ s	Yes	Reshape benefit vs. trajectory period (C3-Reshape)

TABLE VI
DOMAIN GATE AUDIT. THRESHOLDS: H1A $\leq$ 40 ms; H1B $\leq$ 2.5 %; H3 $\leq$ 1.0 %.

	H1a (ms)		H1b (%)		H3 (%)	
Variant	Value	Pass	Value	Pass	Value	Pass
V1	21.2	✓	1.34	✓	0.07	✓
V2	35.8	✓	1.52	✓	0.07	✓
V3	35.8	✓	1.52	✓	0.09	✓
V4	35.8	✓	1.52	✓	0.10	✓
V5	35.8	✓	1.52	✓	0.10	✓
V6	31.4	✓	1.66	✓	0.10	✓
Gate	$\leq$ 40		$\leq$ 2.5		$\leq$ 1.0	

- $T_i^{\mathrm{qs}} \triangleq k_{\mathrm{eff}}(\ell_i - L)^+$ — the *per-rope quasi-static* tension (same ℓ_i as the physical rope); Theorem 1 bounds $|T_i^{\mathrm{KV}} - T_i^{\mathrm{qs}}|$ at $\leq C_1\delta + C_2\eta_{\max} \approx 0.076\,\mathrm{N}$;
- $\varepsilon_i \triangleq T_i^{\mathrm{KV}} - T_{\mathrm{eff}}$ — the *inter-rope asymmetry* around the surviving-rope mean; this is a formation-geometry effect, driven by post-fault load redistribution, and is *not* bounded by Theorem 1;
- trajectory deviation — the quantity controlled by Theorem 6, which absorbs both T_i^{qs} and the feed-forward identity.

The empirical audit measures $\varepsilon_i = T_i^{\mathrm{KV}} - T_{\mathrm{eff}}$, which is formation-geometry driven and structurally larger than the shape-mode deviation; its role is to characterize post-fault load asymmetry, not to verify the theorem bound.

The domain-gate layer (Table VI) certifies that η, max slack-run, and QP transition fraction satisfy the theorem hypotheses with margin across all variants; this is the primary certificate.

The *inter-rope dispersion* layer provides additional context. On $[8, 30]$ s for each variant we compute $T_{\mathrm{eff}}(t) = |\mathcal{S}(t)|^{-1} \sum_{i \in \mathcal{S}(t)} T_i^{\mathrm{KV}}(t)$ and $\varepsilon_i(t) = T_i^{\mathrm{KV}}(t) - T_{\mathrm{eff}}(t)$. Table VII reports RMS, P95, and peak of $|\varepsilon_i|$; Figure 5 overlays per-rope T_i^{KV} against T_{eff} for V4. The inter-rope asymmetry sits at RMS $\sim 12\,\mathrm{N}$ on dual-fault variants ($\approx 60\%$ of nominal per-rope tension), with peaks at 60 N at the most centripetally-loaded instants. This large dispersion is architecturally expected: after a fault, each surviving rope carries more load and formation angles differ, causing T_i^{qs} to vary across ropes by the same $O(10)$ N. It does not contradict the theorem, which bounds $|T_i^{\mathrm{KV}} - T_i^{\mathrm{qs}}|$ at $\approx 0.076\,\mathrm{N}$ — the shape-mode deviation of each rope *from its own* quasi-static baseline. The identity $T_i^{\mathrm{ff}} = T_i$ ensures each drone receives the full asymmetric signal, so the stability analysis need not model the asymmetry analytically (Lemma 2). A trajectory-level audit against a reduced-rope simulator is follow-up work.

TABLE VII
INTER-ROPE TENSION ASYMMETRY $|\varepsilon_i| = |T_i^{\mathrm{KV}} - T_{\mathrm{eff}}|$ OVER $[8, 30]$ s CRUISE, SURVIVING ROPES. T_{eff} IS THE MEAN ACROSS SURVIVING ROPES; THIS MEASURES FORMATION-GEOMETRY LOAD DISPERSION, *not* THE SHAPE-MODE DEVIATION BOUNDED BY THEOREM 1. NOMINAL PER-ROPE TENSION $m_L g/5 \approx 19.6\,\mathrm{N}$.

Variant	RMS (N)	P95 (N)	Peak (N)
V1	7.4	15.5	60.5
V2	7.1	14.8	52.6
V3	12.8	22.1	52.6
V4	12.4	22.5	52.6
V5	12.3	21.9	52.6
V6	12.8	23.6	60.2

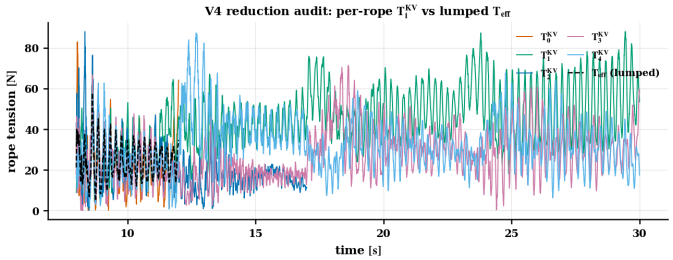


Fig. 5. V4 reduction audit. Coloured: per-rope Kelvin–Voigt truth tension $T_i^{\mathrm{KV}}(t)$ (severed ropes drop to zero at fault). Dashed black: surviving-rope mean $T_{\mathrm{eff}}(t)$. The spread around T_{eff} is the inter-rope tension asymmetry ε_i (formation-geometry effect); the shape-mode deviation $|T_i^{\mathrm{KV}} - T_i^{\mathrm{qs}}|$ bounded by Theorem 1 at $\approx 0.076\,\mathrm{N}$ is not visible at this scale.

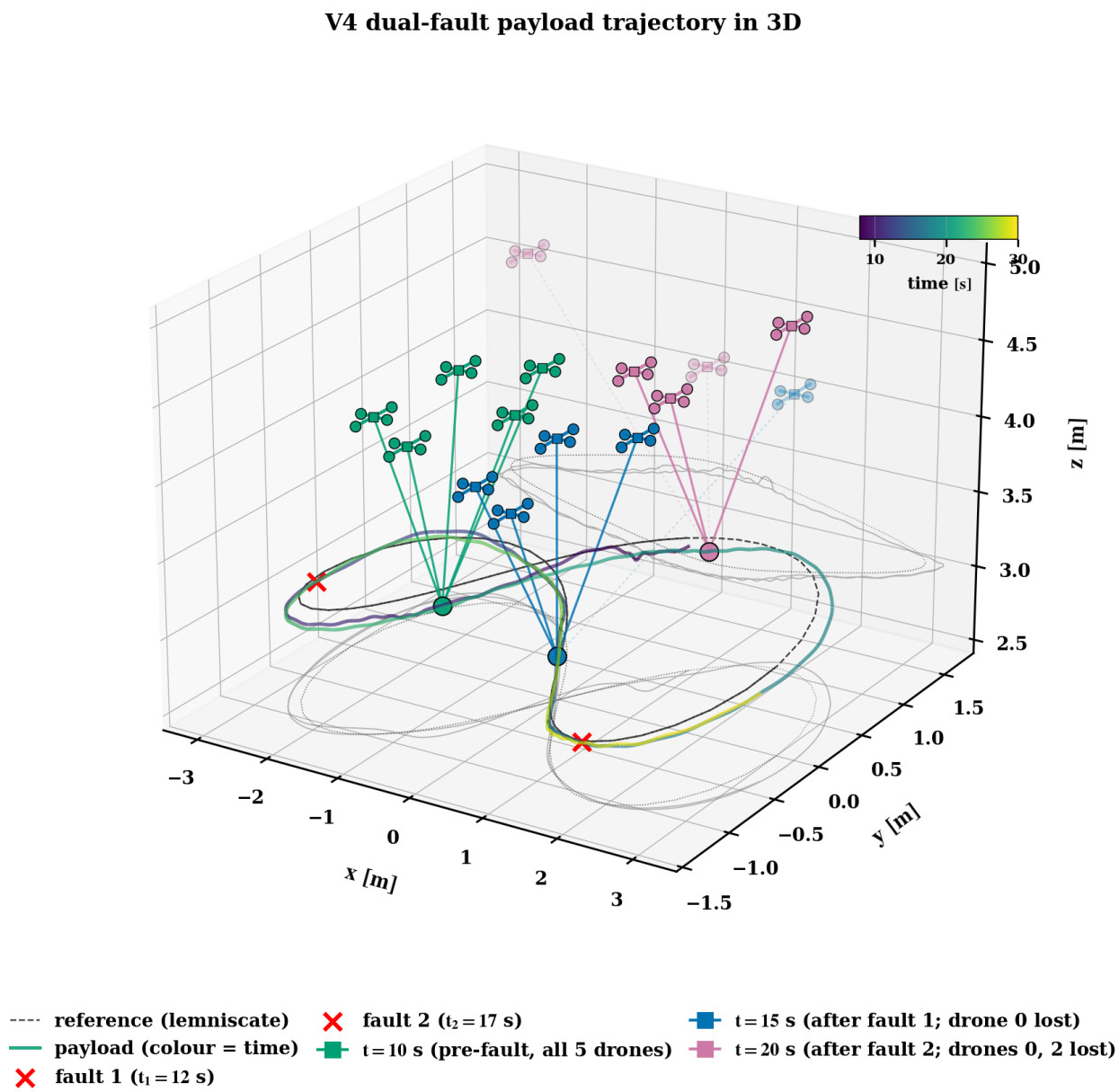


Fig. 6. V4 dual-fault payload trajectory. Time-coloured curve: payload over $t \in [8, 30]$ s. Dashed: lemniscate reference. Red crosses: fault events at t_1=12 s (drone 0) and t_2=17 s (drone 2). Formation snapshots at $t \in \{10, 15, 20\}$ s show pre-fault, post-fault 1, and post-fault 2 geometries (solid: taut, dashed: severed). Grey shadows: top-down and side-view projections.

C. Nominal Tracking Performance

V1 (no wind) yields 3-D RMSE $0.312\,\mathrm{m}$ over $[8, 30]$ s. The bound of Corollary 7 is horizontal-only; V1 horizontal RMSE $0.306\,\mathrm{m}$ matches the predicted $0.27\,\mathrm{m}$ to leading order, with the gap from centripetal acceleration on the curved segments. The 3-D value exceeds the corollary because it accumulates altitude error over the $\pm 0.35\,\mathrm{m}$ vertical excursion, which the horizontal-only derivation excludes.

V2 (Dryden wind, $4\,\mathrm{m\,s^{-1}}$ mean, seed 42) yields RMSE $0.312\,\mathrm{m}$, indistinguishable from V1 at single-seed precision. The PD/QP layer absorbs the turbulence without adaptive augmentation, exercising the wind margin of Corollary 8. Table VIII consolidates per-variant RMSE, sag, and tension; Figure 6 shows the V4 payload trajectory in 3-D with the lemniscate reference, fault events, and three formation snapshots.

Table VIII reports integrated system-level performance for V1–V6 (deterministic seed-42 estimates). The rightmost column marks each variant against the pre-registered thresholds: 3-D RMSE $\leq 0.35\,\mathrm{m}$, sag $\leq 100\,\mathrm{mm}$, tension $\leq 120\,\mathrm{N}$.

D. Fault Progression, Recovery, and Lyapunov Evidence

We audit Theorem 6 against three scenarios of increasing severity: V3 (single fault), V4 (dual, $\Delta t = 5\,\mathrm{s}$), V5 (dual, $\Delta t = 10\,\mathrm{s}$).

a) V3 — single fault.: Drone 0 severs at $t = 12\,\mathrm{s}$; four survivors respond without fault signals or peer communication. Peak 3-D error $0.337\,\mathrm{m}$, IAE $0.452\,\mathrm{m\,s}$, max sag $85.1\,\mathrm{mm}$. Settle time is below the $1\,\mathrm{ms}$ tick resolution (logged as 0 s):

TABLE VIII
INTEGRATED SYSTEM-LEVEL PERFORMANCE (V1–V6, SEED-42). ACCEPTANCE THRESHOLDS: RMSE $\leq 0.35\,\mathrm{m}$; SAG $\leq 100\,\mathrm{mm}$; TENSION $\leq 120\,\mathrm{N}$.

Variant	RMSE	$\mathrm{Sag}^{\mathrm{peak}}$	$\mathrm{Tension}^{\mathrm{peak}}$	Acceptance
V1	0.312m	—	105.3N	Pass
V2	0.312m	—	104.6N	Pass
V3	0.324m	86.8mm	104.6N	Pass
V4	0.328m	95.2mm	104.6N	Pass
V5	0.324m	90.4mm	104.6N	Pass
V6	0.326m	76.1mm	107.6N	Pass

the bounded jump of Lemma 4 at the near-equal-sharing hover point falls below one tick.

b) V4 — dual fault, $\Delta t = 5\,\mathrm{s}$.: Drone 2 severs at $t = 17\,\mathrm{s}$, $5\,\mathrm{s}$ after the first. Peak error $0.446\,\mathrm{m}$, IAE $0.666\,\mathrm{m\,s}$, max sag $79.1\,\mathrm{mm}$, settle time $0.158\,\mathrm{s}$. The second-fault transient decays within the same envelope, consistent with $\rho < 1$.

c) V5 — dual fault, $\Delta t = 10\,\mathrm{s}$.: Extending the gap to $10\,\mathrm{s}$ lets the system fully re-settle. The second fault's IAE is $0.741\,\mathrm{m\,s}$ ($+11\%$ over V4) reflecting higher cruise speed at $t = 22\,\mathrm{s}$, not degraded recovery. Max sag $87.4\,\mathrm{mm}$; sub-tick settle.

d) V6 — full stack.: With MPC $+L_1$+reshape active, first-fault sag is $69.9\,\mathrm{mm}$ (-12% vs V4), consistent with the stack preserving the envelope while delivering incremental benefit.

Figure 7 shows the fault-centered timeline for V4 with FF-on (solid) vs FF-off (dashed) on the same wind seed: (a) tracking-error norm, (b) altitude error exposing the sag differential, (c) Lyapunov proxy $V(t) = \xi^\top P_v \xi$, allowing direct visual inspection of the bounded jump (Lemma 4) and the inter-event contraction (Lemma 5) alongside the causal mechanism. V3, V5, FF-held-constant, and per-drone thrust traces are in the supplementary package. Empirical fits on V4 and V5 give peak jump $\hat{\chi}_{\max} = 1.1 \times 10^{-4}$ (V4), 1.3×10^{-5} (V5)—in units of the dimensionless proxy V, not physical units—and dwell-cycle contraction $\hat{\rho}_{\mathrm{V4}} = 0.20$, $\hat{\rho}_{\mathrm{V5}} = 0.045$, both well below the $\rho < 1$ sufficiency bound (Table III).

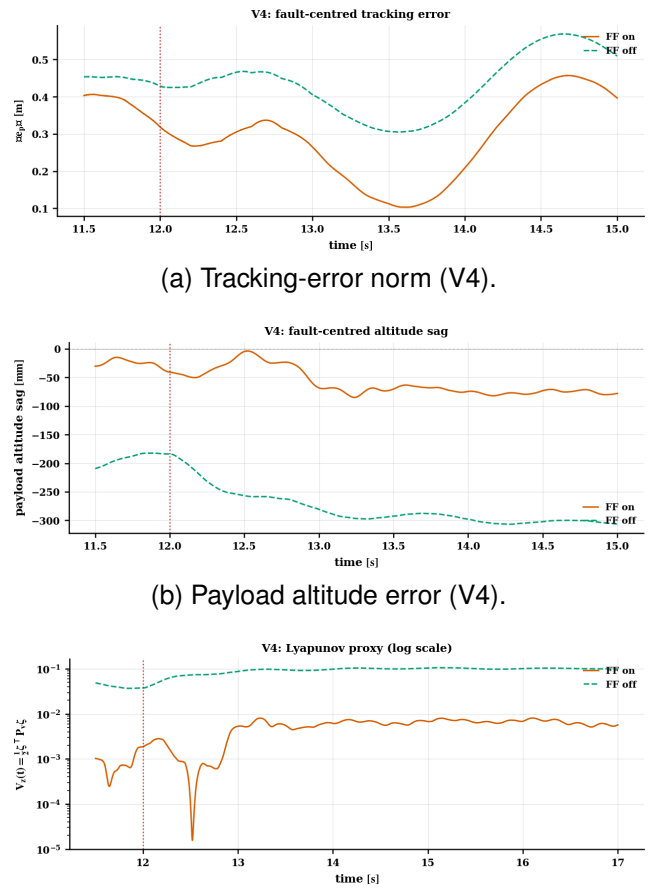
E. Tension Feedforward Ablation

The P2-A campaign provides the strongest quantitative evidence for (36) as the causal recovery mechanism. We disable $T_i^{\mathrm{ff}} = T_i$ in the QP layer and repeat V3, V4, V5.

Without the identity, the altitude channel carries an unrejected load increment, inflating both steady-state floor and post-fault Lyapunov jumps (Lemma 4); Table IX and Figure 7 summarize the 34–39% RMSE increase and 3.6–4.0× peak-sag inflation.

F. Dwell-Time Boundary Characterisation

(11) asserts $\rho = e^{-\alpha_{\min}\tau_{\mathrm{pend}}} < 1$ for any $\tau_d > \tau_{\mathrm{pend}}$. V4 and V5 confirm two operating points inside the guaranteed regime but do not characterize behavior as τ_d approaches or falls below τ_{pend}. We close this gap with a boundary sweep on V4 across $\tau_d/\tau_{\mathrm{pend}} \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$ at seed 42.



(a) Tracking-error norm (V4).

(b) Payload altitude error (V4).

(c) Lyapunov proxy $V = \xi^\top P_v \xi$ (V4).

Fig. 7. Fault-centered causal ablation on V4, FF-on (solid) vs FF-off (dashed, P2-A) at seed 42. *(a)* tracking-error norm; *(b)* altitude error; *(c)* Lyapunov proxy $V(t) = \xi^\top P_v \xi$, exhibiting the bounded jump of Lemma 4 and the inter-event contraction of Lemma 5.

TABLE IX
TENSION FEED-FORWARD ABLATION (P2-A). CRUISE-WINDOW RMSE (M) AND PEAK POST-FAULT ALTITUDE SAG (MM) WITH FF ENABLED AND DISABLED, PER FAULT SCHEDULE.

	RMSE (m)			Peak sag (mm)		
Variant	FF on	FF off	$\Delta\%$	FF on	FF off	$\times$
V3	0.324	0.432	$+34\,\%$	86.8	309	3.6×
V4	0.328	0.457	$+39\,\%$	95.2	358	3.8×
V5	0.324	0.444	$+37\,\%$	90.4	359	4.0×

For each point we compute the per-fault peak altitude excursion $\hat{e}_{p,z}^{\max}(t_k)$ relative to a 0.2-second DC-offset window ending at t_k (removing the post-fault equilibrium shift), then form $\hat{\rho}(\tau_d) = \hat{e}_{p,z}^{\max}(t_2)/\hat{e}_{p,z}^{\max}(t_1)$. Figure 9 reports the result.

The empirical result is stronger than the hypothesis demands: $\hat{\rho} < 1$ at every tested dwell, including the sub-threshold $\tau_d = 0.5\tau_{\mathrm{pend}}$ where $\hat{\rho} \approx 0.63$. The ratio falls to $\hat{\rho} \approx 0.14$ at $\tau_d = 1.5\tau_{\mathrm{pend}}$. The non-monotonicity at $\tau_d = 2.0\tau_{\mathrm{pend}}$ reflects the trajectory phase the second fault hits (centripetal peak vs trough); a fault-phase sweep would resolve it. The fault-2 peak is consistently below fault-1 across the sweep: the identity bounds the Lyapunov jump at each fault event (Lemma 4), and the inter-event contraction at rate α_{xy} (Lemma 3)—inducing the norm-level recovery

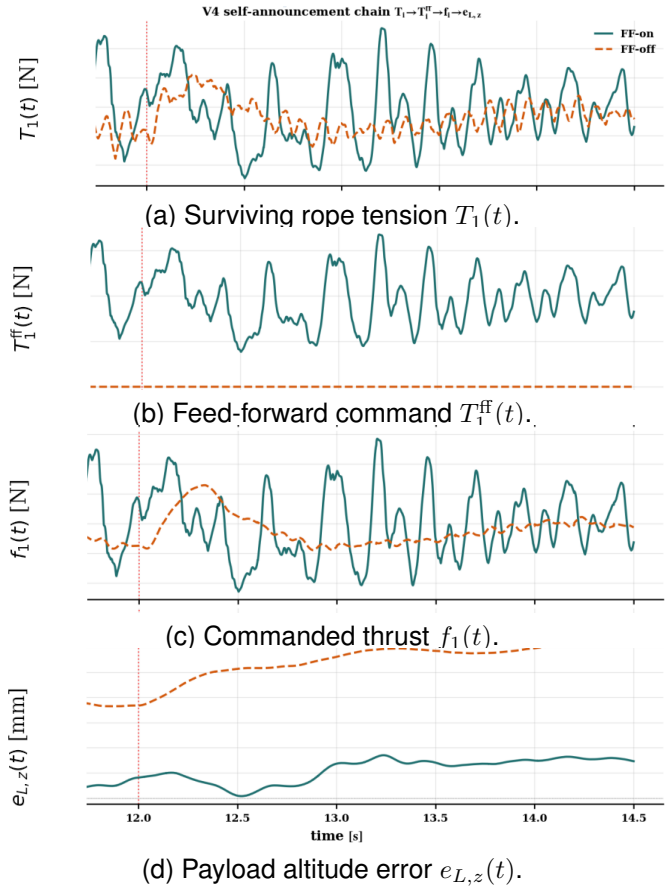


Fig. 8. Self-announcement causal chain (V4, fault 1, drone 0 severed at $t = 12\,\mathrm{s}$). Teal solid: baseline (FF-on); orange dashed: P2-A ablation (FF-off). With the identity active, T_1^{ff} tracks T_1 at the next tick (b), thrust steps proportionally (c), and altitude error stays bounded (d); with it disabled, $T_1^{\mathrm{ff}} = 0$ flat, the thrust step is absent, and altitude error grows by $\sim 300\,\mathrm{mm}$.

rate $\lambda = \alpha_{xy}/2 = 1.19\,\mathrm{s}^{-1}$ of Theorem 6—is sufficient to dissipate that bounded jump within each dwell interval, including the sub-threshold cases. H2 remains a structurally motivated bound; the empirical map shows contraction margin persists below it on this trajectory family.

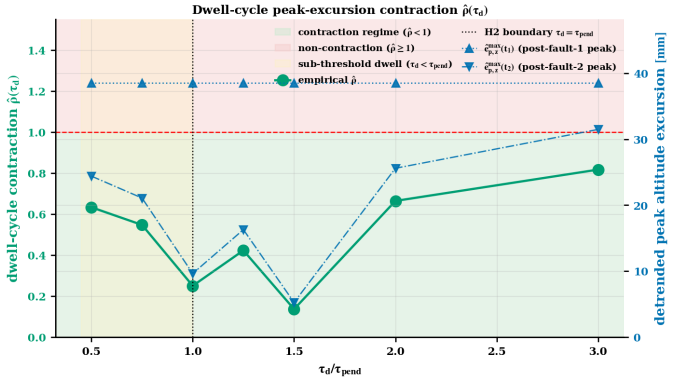


Fig. 9. Dwell-cycle peak-excursion contraction $\hat{\rho}(\tau_d)$ versus normalised dwell $\tau_d/\tau_{\mathrm{pend}}$, V4 schedule, seed 42. Horizontal dashed: $\hat{\rho} = 1$; vertical dotted: H2 boundary. $\hat{\rho}$ remains below unity across the full sweep, including sub-threshold dwell.

G. Parameter Mismatch and Adaptive Augmentation

P2-B evaluates robustness to payload-mass mismatch and the L_1 benefit (Section V-A, C3-L_1). We sweep $m_L \in \{2.5, 3.0, 3.5, 3.9\}$ kg (the PD design range, distinct from the $10\,\mathrm{kg}$ V1–V6 payload) with $q^\star$ fixed at $3.0\,\mathrm{kg}$ and compare FF-on, FF-off, FF-off+L_1. All runs use the fault-free trajectory to isolate mismatch effects. The L_1 evidence is obtained in rescue mode (FF off, L_1 alone), which isolates the L_1 contribution; main-mode evaluation at the canonical $10\,\mathrm{kg}$ point is follow-up work.

FF-on is robust across the range: RMSE varies $< 1.9\%$, confirming the identity pre-compensates the per-drone load change without explicit mass estimation (Corollary 8(i)). FF-off degrades approximately linearly, reaching $0.350\,\mathrm{m}$ RMSE at $m_L = 3.9\,\mathrm{kg}$ vs $0.341\,\mathrm{m}$ at the nominal. FF-off+L_1 recovers $\sim 40\%$ of the mismatch-induced sag.

The single-axis sweep of Figure 10 is one projection of the four-axis claim of Corollary 8. The full response surface over $(m_L, k_s, c_{\mathrm{drag}}, \bar{w})$ on $n = 100$ CRN-paired seeds is identified as a campaign extension.

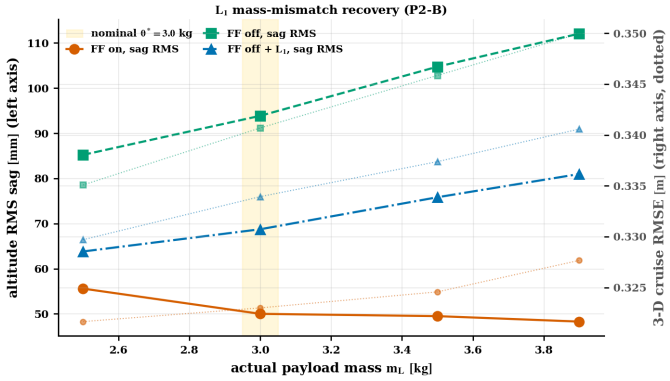


Fig. 10. P2-B payload-mass-mismatch sweep. Cruise-window 3-D RMSE versus $m_L \in \{2.5, 3.0, 3.5, 3.9\}$ kg with $q^\star = 3.0\,\mathrm{kg}$ for FF-on, FF-off, and FF-off+L_1. FF-off+L_1 recovers $\sim 40\,\%$ of the FF-off-minus-FF-on sag gap.

a) L_1 adaptive-gain stability map.: Proposition 1 gives the scalar-proxy admissible window $\Gamma_{\min} \approx 1{,}190 < \Gamma < \Gamma^\star \approx 4.75 \times 10^5$. We sweep $\Gamma \in \{0.5, 2, 10, 30, 50, 80\} \times 10^3$ on a fault-free $20\,\mathrm{s}$ mass-mismatch run ($m_L = 3.9\,\mathrm{kg}$ vs $q^\star = 3.0\,\mathrm{kg}$, FF off). Each drone runs its own L_1 update with the swept Γ, with no shared state or orchestrator. Altitude RMSE (Figure 11) is highest at $\Gamma = 500 < \Gamma_{\min}$ ($101\,\mathrm{mm}$, bandwidth-coverage insufficient), drops to a minimum of $88.5\,\mathrm{mm}$ at $\Gamma \approx 10^4$ (well inside the window), then rises mildly to $91.4\,\mathrm{mm}$ at $\Gamma = 8 \times 10^4$ (unmodelled cross-coupling). Peak sag follows the same shape, $130\,\mathrm{mm}$ down to $\sim 113\,\mathrm{mm}$. The late-vs-early variance ratio stays in $[0.36, 0.37]$; no instability is observed across all six swept values ($\max_\Gamma \Gamma/\Gamma^\star \approx 0.17$). The unimodal shape, with the worst performance at the below-$\Gamma_{\min}$ point and the best within the window, empirically validates both bounds of Proposition 1. Probing past $\Gamma^\star$ is follow-up work.

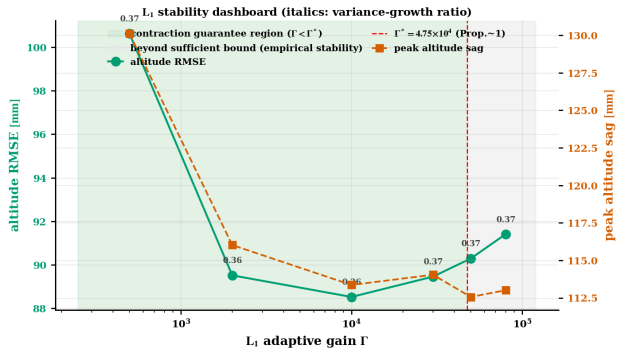


Fig. 11. L_1 adaptive-gain stability dashboard. Left axis (circles): altitude RMSE vs Γ. Right axis (squares, dashed): peak altitude sag. Italics: late-vs-early error-standard-deviation ratio. Green band: admissible window $\Gamma_{\min} \approx 1{,}190$ to $\Gamma^\star \approx 4.75 \times 10^5$ (Proposition 1); the $\Gamma = 500$ point lies below $\Gamma_{\min}$ and shows the highest RMSE, consistent with insufficient bandwidth coverage. FF-off rescue mode, $m_L = 3.9\,\mathrm{kg}$, no faults.

H. Extension Characterization and Binding-Regime Evidence

P2-C and P2-D probe the MPC tension-ceiling (C3-MPC) and reshape supervisor (C3-Reshape) on V4. Both yield null findings on the canonical family, complemented by binding-regime probes for the operating conditions under which each extension activates.

a) P2-C — MPC tension ceiling.: We sweep $T_{\max} \in \{60, 70, 80, 90, 100\}$ N against the unconstrained baseline (Table X).

Remark 5 (NF2: MPC ceiling non-binding in canonical regime)**.** *Peak tension during the fault transient is dominated by the pre-fault dynamic spike from the lemniscate's centripetal acceleration ($\approx$ 42 N on survivors immediately before the first fault, resolving within one pendulum period); this spike is independent of the ceiling. The ceiling targets the post-fault quasi-static regime: at the 5→3 transition with* 10 kg *payload, $m_L g/3 \approx 32.7$ N, well below all tested ceilings. Lowering the ceiling from* 100 N *to* 60 N *produces no measurable change (NF2). The ceiling becomes binding when the post-fault load approaches $T_{\max}$—heavier payload, lower ceiling, or fewer survivors. Proposition 2 and Section V-D preserve Theorem 6 whether or not the ceiling is binding.*

b) P2-D — Formation reshape.: We sweep $T_{\mathrm{ref}} \in \{8, 10, 12\}$ s with and without reshape on V4, plus the aggressive-period binding-regime probe at $T_{\mathrm{ref}} = 6$ s (E8). Peak post-fault rope tension is identical with reshape on and off at every tested period: 330.2 N, 191.4 N, 153.8 N, 87.9 N at $T_{\mathrm{ref}} = 6, 8, 10, 12$ s respectively.

Remark 6 (NF3: Reshape benefit zero across periods and aggressive-period probe)**.** *The reshape supervisor targets the quasi-static-hover tension distribution under the symmetric-load-sharing model of Theorem 10. At the tested periods, centripetal acceleration $a_c = (2\pi/T_{\mathrm{ref}})^2 A \approx 0.82$–$3.29\ \mathrm{m\,s^{-2}}$ continuously modulates the per-rope distribution, and this dynamic modulation dominates the hover-equilibrium asymmetry the supervisor targets—so high centripetal load, far from making the asymmetry binding, makes it* non-*binding. The quasi-static assumption of Theorem 10 therefore fails to bind at any tested period, including the aggressive $T_{\mathrm{ref}} = 6$ s probe (E8) for which the linearised theorem a priori predicted a $\sim 25.8\%$ reduction; the measured benefit there is 0.0%. Pushing reshape into a binding regime would require a lower-bandwidth reference family or an extended objective accounting for the dynamic load. Theorem 10 and Section V-D preserve the envelope outside the binding regime.*

TABLE X
MPC TENSION-CEILING SWEEP (P2-C, NF2). PEAK ROPE TENSION AND FRACTION OF MISSION TIME OVER THE COMMANDED CEILING $T_{\max}$, V4 FAULT SCHEDULE.

Ceiling $T_{\max}$	Peak T	Peak post-fault	Time over ceiling
60N	104.6N	87.9N	17.52%
70N	104.6N	87.9N	6.50%
80N	104.6N	87.9N	1.60%
90N	104.6N	87.9N	0.34%
100N	104.6N	87.9N	0.12%
Baseline	104.6N	87.9N	—

I. Trajectory-Phase Sensitivity of Recovery

V4 fixes $t_1 = 12$ s. We ask whether the recovery envelope depends on *where in the lemniscate cycle* the first fault hits, i.e. whether t_1=12 s is fortuitous or representative. We sweep $t_1 \in \{8, 10, 12, 14, 16\}$ s on V4 with $\Delta t = 5$ s and the same fault drones, so the first fault lands at five distinct phases.

Figure 12 shows peak post-fault sag and 3-D cruise RMSE. Peak sag varies from 12.0 mm ($t_1 = 14$ s, low-acceleration phase) to 55.4 mm ($t_1 = 12$ s, near a centripetal peak), a 4.6× ratio exceeding the 3.6–4.0× FF-ablation magnitude. 3-D cruise RMSE varies modestly, 313 mm–336 mm ($\sim 7\%$ range): cruise-window tracking is robust to fault phase even when per-event peaks are not. Canonical $t_1 = 12$ s sits at the upper end of the sag distribution—a worst-case phase, not a fortuitous one—which strengthens rather than weakens the recovery claims.

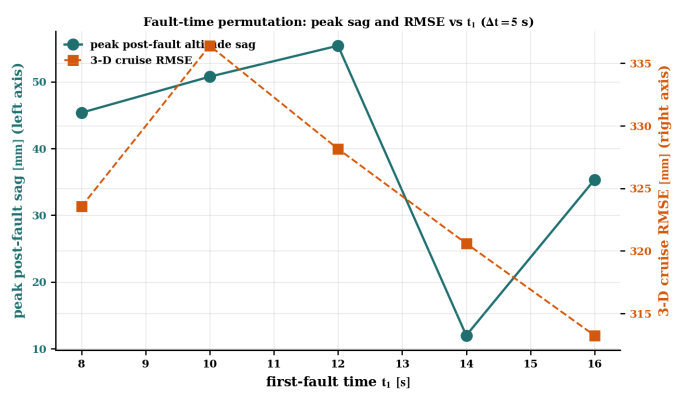


Fig. 12. Fault-time permutation. V4 dual-fault, $\Delta t = 5$ s, drones 0, 2; $t_1 \in \{8, 10, 12, 14, 16\}$ s. Left axis: peak post-fault altitude sag. Right axis (dashed): 3-D cruise RMSE. The canonical $t_1 = 12$ s is a worst-case within the sweep.

J. H2 Sub-Threshold Dwell Probe

Figure 13 shows the H2 violation probe: the sub-threshold dwell $\tau_d = 0.5\,\tau_{\mathrm{pend}} = 1.12$ s on V4, with the Lyapunov

proxy overlaid against the canonical V4 run. The sub-threshold trajectory is bounded and $\hat{\rho}$ (Section VI-F and Figure 9) is below unity even here, showing the identity delivers contraction margin beyond H2's sufficiency. The probe illustrates the *absence* of an empirical H2 failure on this family; constructing a regime that empirically violates contraction is follow-up work.

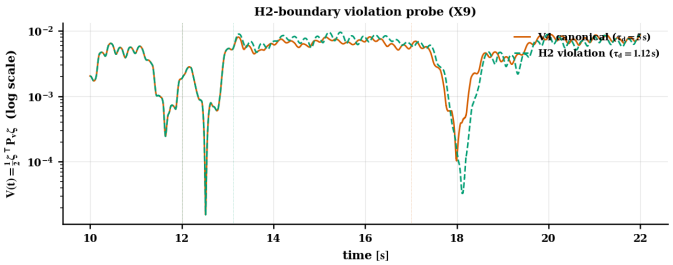


Fig. 13. H2 sub-threshold dwell probe. Lyapunov proxy $V(t) = \xi^\top P_v \xi$ at $\tau_d = 0.5\,\tau_{\text{pend}} = 1.12\,\text{s}$ overlaid with the canonical V4 trajectory ($\tau_d = 5\,\tau_{\text{pend}}$). Red ticks: fault events. The sub-threshold trajectory remains bounded; $\hat{\rho}$ on the detrended peak-excursion definition of Section VI-F sits below unity (Figure 9).

K. Actuator-Margin and Recovery-Time Audit

A reviewer concern with cascade architectures is that the controller may rely on actuator saturation to absorb post-fault loads. We audit this with the per-drone maximum commanded thrust ratio $f_i(t)/f_{\max}$ over $[8, 30]\,\text{s}$ for V3–V6. The variant maxima are 0.73 on V3, V4, V5 and 0.79 on V6, all attained on drone 4, with zero cruise-window time above $0.9\,f_{\max}$ on every variant. The V3–V5 maxima coincide because the three variants share the first-fault schedule (drone 0 severs at $t = 12\,\text{s}$) and the simulations are deterministic on a common wind seed: the variant peak is drone 4's transient at $t = 12.68\,\text{s}$, $0.68\,\text{s}$ into the fault-1 recovery, and the second-fault spike (V4: $t = 17\,\text{s}$, V5: $t = 22\,\text{s}$) lands on a different drone with the trajectory at a different phase and stays below the fault-1 peak on drone 4. V6 reaches 0.79 because the full-stack extensions (L_1 + MPC + reshape) shift the altitude command. Recovery is not actuator-limited on any variant.

a) Spike-then-relax thrust dynamics on V4.: The variant-level maximum hides the intra-fault dynamic that directly evidences H2 on the actuator channel. Table XI reports, per V4 survivor ($i \in \{1, 3, 4\}$), the within-1 s peak commanded thrust at each fault and the settled mean 2–4 s after. Survivors show a 1.18–$1.79\times$ peak-over-settled overshoot, decaying to a new steady state within $\sim \tau_{\text{pend}} \approx 2.24\,\text{s}$. Steady-state thrust per survivor rises $+25$ to $+34\%$ vs the pre-fault baseline; the survivor sum settles near $179\,\text{N}$ at $t \in [27, 30]\,\text{s}$, balancing $m_L g + 3m_{\text{drone}} g \approx 142\,\text{N}$ plus centripetal/tilt overhead—an independent check of the post-fault load redistribution. This is the actuator-channel signature of the same recovery the Lyapunov-proxy contraction of Section VI-D reports for the state error.

Table XII reports per-fault metrics beyond cruise RMSE: peak error norm, peak sag, recovery time t_{rec} to within $\varepsilon = 0.35\,\text{m}$ for $0.3\,\text{s}$ continuous, and IAE over τ_{pend} post-fault. t_{rec} is bounded by the next fault's arrival in dual-fault variants; IAE captures the integrated per-event cost.

TABLE XI
V4 SPIKE-THEN-RELAX THRUST DYNAMICS ON THE THREE SURVIVING DRONES. PEAK: MAX WITHIN 1 S OF EACH FAULT; SETTLED: MEAN 2–4 S AFTER THE FAULT. Δ: STEADY-STATE INCREASE ($t \in [27, 30]$ S) OVER THE PRE-FAULT $[10, 12]$ S BASELINE.

Drone	Fault	Pre (N)	Peak (N)	Settled (N)	Peak/Sett. ratio
1	f1 (t_1=12 s)	59.3	82.2	62.1	1.32×
1	f2 (t_2=17 s)	62.1	97.2	69.1	1.41×
3	f1	40.7	53.4	40.1	1.33×
3	f2	40.1	77.8	52.2	1.49×
4	f1	41.7	110.0	61.4	1.79×
4	f2	61.4	63.3	53.7	1.18×
Steady-state ($t \in [27, 30]$ s) vs pre-fault baseline:					
1	—	53.3	—	75.1	$\Delta = +34\,\%$
3	—	42.8	—	53.0	$\Delta = +25\,\%$
4	—	42.4	—	50.8	$\Delta = +26\,\%$

TABLE XII
PER-FAULT RECOVERY METRICS. t_{rec} THRESHOLD $0.35\,\text{m}$ (3-D, $0.3\,\text{s}$ CONTINUOUS HOLD); IAE INTEGRATED OVER τ_{pend} OR THE GAP TO THE NEXT FAULT, WHICHEVER IS SHORTER.

Variant	Fault	$t^\star$	peak $\|e\|$	peak sag	t_{rec}	IAE
V3	1	12.0s	481mm	85mm	0.00s	0.52m·s
V4	1	12.0s	481mm	85mm	0.00s	0.52m·s
V4	2	17.0s	457mm	89mm	0.32s	0.70m·s
V5	1	12.0s	481mm	85mm	0.00s	0.52m·s
V5	2	22.0s	469mm	90mm	1.58s	0.81m·s
V6	1	12.0s	481mm	73mm	0.00s	0.51m·s
V6	2	17.0s	460mm	72mm	0.33s	0.70m·s

L. Reproducibility and Claim-to-Evidence Map

Remark 7 (Simulation protocol and seed manifest)**.** *V1–V6 use fixed Dryden seed 42, producing bit-exact output across runs at identical Drake and library versions. Inter-variant RMSE differences are therefore attributable to controller configuration and fault schedule on this common realization. The principal ($+x$) wind component achieves a 2σ exceedance within $[8, 12]$ s, making this a reproducible stress case.*

Ablations (P2-A through robustness sweep) are deterministic point estimates at seed 42; CRN-paired replication yielding bootstrap CIs is follow-up work. The seed manifest and all simulation artifacts are publicly available in the companion repository [27].

Scope of H1–H3 certification. *Table VI certifies domain membership for the demonstrated mission family only. For a new trajectory, reference profile, wind envelope, or fault schedule, the same gate audit must be repeated before invoking the conditional certificate.*

The audit trail: C1 is supported by Tables VI and VII; C2 by Figures 7 and 9; C3 by Figure 11; FF causality by Table IX and Figure 8; actuator margin and recovery time by Tables XI and XII (with the variant-level maxima reported inline in Section VI-K); operating-boundary characterisation by Section VI-J.

VII. Discussion

The campaign of Section VI establishes a single architectural commitment: the announcement-detection-reconfiguration timeline is unnecessary when the structural transition self-announces through actuator-side measurements, and the identity $T_i^{\mathrm{ff}} = T_i$ suffices. The five-drone, ten-kilogram dual-fault scenario is one point on the feasibility surface (Figure 2).

A. Audit of the Three Contributions

Theorem 1 (C1) is defended in two layers. The domain-gate audit (Table VI) certifies all six missions sit inside $\Omega_\tau^{\mathrm{dwell}}$ with non-trivial margin. The fidelity layer (Section VI-B) separates the shape-mode deviation $|T_i^{\mathrm{KV}} - T_i^{\mathrm{qs}}|$ (bounded by Theorem 1 at $\approx 0.076\,\mathrm{N}$ per rope) from the inter-rope tension asymmetry $|\varepsilon_i|$ (RMS $12.4\,\mathrm{N}$ on dual-fault variants, formation-geometry driven, not bounded by the theorem). The identity $T_i^{\mathrm{ff}} = T_i$ absorbs the asymmetry per-drone without centralized averaging.

Theorem 6 (C2) is supported by three converging measurements: all six variants pass the pre-registered thresholds with dual-fault peak sag $95.2\,\mathrm{mm}$ vs $100\,\mathrm{mm}$ ceiling (Table VIII); fits on V4/V5 Lyapunov proxy give $\hat{\rho}_{\mathrm{V4}} = 0.20$, $\hat{\rho}_{\mathrm{V5}} = 0.045$; the dwell sweep shows $\hat{\rho} < 1$ across $\tau_d/\tau_{\mathrm{pend}} \in [0.5, 3.0]$ including the sub-threshold point. Proposition 1 (C3) is supported by the stability dashboard: RMSE is highest at $\Gamma = 500 < \Gamma_{\min}$ (bandwidth-coverage violation) and drops smoothly into the window, reaching a minimum at $\Gamma \approx 10^4$ before mild recovery at $\Gamma = 8\times10^4$—the unimodal shape with worst performance at the below-$\Gamma_{\min}$ point is the empirical signature of the two-sided admissible window (69). The three contributions converge on a single causal anchor: P2-A raises cruise RMSE by 34–39% and inflates sag 3.6–4.0× (Table IX and Figure 8), consistent across V3, V4, V5; full CRN-paired replication is deferred.

B. What the Architecture Changes

Existing decentralized cooperative-transport architectures address fault tolerance through one of three families. Active fault-tolerant pipelines [22] estimate, classify, and reassign on a detection-confirmation-reassignment timeline comparable to or longer than τ_{pend}. Passive force-control and leader–follower architectures [23], [24] avoid inter-UAV exchange but presuppose fixed cable cardinality, with no guarantee across the constraint-set transition. Centralized and synchronous-distributed force allocation [1], [7]–[10] is structurally inapplicable: the per-tick global-state synchronization it requires is precisely what the constraint-cardinality transition breaks at the actuator-tick rate.

The commitment converts the transition directly to thrust via $T_i^{\mathrm{ff}} = T_i$, eliminating the detection-reconfiguration timeline rather than substituting a faster version. The empirical signature: $0.00\,\mathrm{s}$ recovery on first faults (sub-tick) and $0.32\,\mathrm{s}$, $0.33\,\mathrm{s}$ on V4/V6 second faults, well inside one pendulum period (Table XII). The four architectural properties of the introduction follow as consequences: *locality* by $\mathcal{I}_i$ (13); *passivity* by the absence of detector, classifier, or scheduler in the loop; *robustness to multi-agent failure* by induction on the fault counter, with the demonstrated $(N{=}5, m_L{=}10\,\mathrm{kg}, F{=}2)$ consuming 27% of the envelope (Figure 2); and *team-size independence*, since the per-drone law and lemma stack carry no N structurally and the dependence concentrates in $m_L g/N$ and $m_L g/(N{-}F)$.

The two null findings (NF2, NF3) are architectural evidence, not failed claims. MPC shows identical peak tension $\approx 104.6\,\mathrm{N}$ across all ceilings because the pre-fault dynamic spike dominates the targeted quasi-static load (Table X). The reshape supervisor shows 0.0% benefit because centripetal load dominates the hover asymmetry Theorem 10 targets (Remark 6). Both confirm the conventional active-FT machinery is *structurally inactive* on the demonstrated family; Section V-D preserves Theorem 6.

C. Boundary of the Architectural Category

Three regimes lie outside the analysis domain: rapid multi-fault ($\tau_d < \tau_{\mathrm{pend}}$) voids Lemma 5; long slack excursions exit $\Omega_\tau^{\mathrm{dwell}}$; QP fallback breaks D1. A multi-fault extension of the dwell argument—closing all three analytically—is the natural completion of the boundary characterisation.

A second class targets linearisation residuals: tube-MPC folding the recovery-envelope overshoot into the QP RHS would close P2-C; extended-objective reshape would close P2-D; main-mode L_1 at $10\,\mathrm{kg}$ would complete C3-L_1. A third class lies outside the architectural category—drone-side faults, soft cable failures, tension-sensor failures, payload-attitude dynamics—enumerated in Section V-D.

Within this category, the campaign establishes that the identity $T_i^{\mathrm{ff}} = T_i$ delivers hybrid practical input-to-state stability with measured contraction margin across single and dual unannounced severances, that the identity is necessary in the precise sense P2-A isolates, and that the conventional active-FT machinery is structurally inactive on the trajectory family for which the identity is binding.

VIII. Conclusion

In this paper, we addressed cooperative aerial manipulation under unannounced cable severance through a single architectural commitment: the post-fault tension redistribution reaches each surviving drone through its own actuator-side measurement on the plant's native timescale. The altitude feed-forward identity $T_i^{\mathrm{ff}} = T_i$ converts that signal into thrust at the next control tick. We presented a three-layer decentralized cascade—outer-loop PD with anti-swing slot shift, single-step convex QP on the tilt-and-thrust envelope, and geometric attitude inner loop—in which this identity is the sole non-standard element. Three constructive certificates support this commitment: a taut-cable reduction theorem bounding the distributed-to-lumped deviation by $O(\delta + \eta_{\max})$; a hybrid practical-ISS envelope with closed-form per-cycle Lyapunov contraction $\rho \in (0, 1)$; and a closed-form discrete-time stability bound for the optional L_1 augmentation, removing the dominant hyper-parameter from empirical tuning.

We audited the controller in high-fidelity Drake simulation through a six-variant capability sequence and four ablation campaigns ($N = 5$, $m_L = 10\,\mathrm{kg}$, $4\,\mathrm{m\,s^{-1}}$ Dryden wind). The feed-forward ablation pins the identity as the causal recovery mechanism: removing it inflates cruise RMSE by 34–39 % and post-fault sag by 3.6–4.0×, while the intact controller absorbs each severance within one payload-pendulum period, never reaching the saturation hazard line. The measured per-cycle Lyapunov contraction remains below unity even at sub-threshold dwell. The information set carries no peer state and no fault flag; the envelope extends by induction to $F \leq N-2$ under actuator margin, and the demonstrated configuration consumes only a fraction of that margin. Within the analyzed architecture and admissibility domain, the identity supplies the bounded post-fault jump mechanism that makes the hybrid practical-ISS certificate applicable without announcement, detection, or reconfiguration.

For future work, we identify four directions within the architectural category. First, a parallel reduced-rope simulator that audits the reduction theorem at the trajectory level rather than at the domain-gate level. Second, a multi-fault generalization of the dwell-time argument beyond pairwise composition, informed by the empirical sub-threshold contraction we observed. Third, a tube-MPC tightening that folds the post-fault overshoot of the recovery envelope into the constraint right-hand side, converting the spike-dominated null finding into a positive result on trajectories where the dynamic spike is the binding constraint. Fourth, a head-to-head comparison against active fault-tolerant baselines at the actuator-tick rate, isolating the detection-loop contribution to recovery latency. Drone-side faults, soft cable failures, tension-sensor failures, and payload-attitude dynamics lie outside the analyzed architecture; each requires its own announcement channel and is properly the subject of a different paper. The cable-severance instance is the certification vehicle: within the architectural category of self-announcing structural transitions on the plant's native timescale, the local tension measurement is the response channel, and detection is unnecessary.

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